ENGINEERING TRUTHINESS

A VALIDATION FRAMEWORK FOR SILVER-LABEL EVALUATION IN ML

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On the Origin of "Truthiness"

The term "truthiness" was coined by Stephen Colbert on the premiere episode of The Colbert Report (October 17, 2005) to describe claims that "feel right" without regard for evidence or logic. We adopt the concept deliberately: "engineering truthiness" describes the condition in which machine learning evaluation appears empirically grounded but rests on proxy labels whose validity has not been formally established. This paper provides the engineering discipline to address it systematically.

Abstract

Machine learning increasingly relies on proxy-derived silver labels rather than human-verified gold labels. These proxies are widely used across clinical informatics, information retrieval, NLP, and computer vision, yet no general framework exists for establishing when silver-label evaluation is defensible.

We introduce three contributions: SILVER, a six-dimension standard for documenting proxy labels; Silver Evaluation Constraints (SEC), which prevent evaluation circularity; and the Silver Risk Index (SRI), which quantifies ranking stability under proxy disagreement.

We define engineering truthiness as the condition arising when silver labels are used without such governance.

Five empirical studies demonstrate the framework: synthetic noise simulation shows ranking instability is regime-dependent; IMDB/Snorkel weak supervision shows 0% ranking flip despite massive calibration gap; clinical trial matching (5.27M pairs) reveals sharp threshold sensitivity and subgroup fairness disparities; CIFAR-10 analysis demonstrates error-structure auditing; and external validation against TREC 2021 expert gold judgments yields SRI = 0.00, confirming that silver-based model selection agrees with independent gold evaluation.

1. Introduction

1.1 The Problem

Access to high-quality labeled data has long been a driving force behind advances in machine learning. In many contemporary applications, however, authoritative ground truth is unavailable

at scale, delayed in time, or fundamentally ambiguous even for domain experts. Clinical outcomes may lack definitive adjudication, relevance judgments may be subjective or context-dependent (Plank, 2022), and large-scale annotation may be infeasible due to cost, privacy, or ethical constraints. As a result, modern systems increasingly rely on labels generated through indirect mechanisms, including weak supervision, distant supervision, behavioral logs, similarity heuristics, and automated or semi-automated annotation pipelines (Frénay & Verleysen, 2014; Ratner et al., 2020; Paullada et al., 2021).

Consider an analogy from consumer finance. No one has a "true creditworthiness number" inscribed somewhere in the universe. FICO scores are engineered proxies built from payment history, debt ratios, and credit utilization; all indirect signals, none of which constitutes ground truth. Yet the entire financial system uses them: when a bank says "our lending model identifies low-risk borrowers better than a competitor's," it is ranking models against a proxy label. The question is not whether credit scores are "real"; they are not, in the same sense that gold labels from expert annotation are not perfectly real either, but whether they are well-enough engineered for the decision at hand. This paper asks the same question about silver labels in machine learning and provides the formal tools to answer it. When is a proxy-derived label a defensible basis for model comparison? Under what conditions does evaluation against silver labels produce the same conclusions as evaluation against gold? And when it does not, can the risk be measured and disclosed? The framework we introduce - SILVER, SEC, and SRI- answers each of these questions.

These proxy-derived labels, commonly referred to as silver labels, are not a methodological shortcut but an operational necessity. In clinical informatics, diagnoses are frequently approximated using electronic health record codes or extracted concepts rather than clinician adjudication (Banda et al., 2018; Oellrich et al., 2016). In information retrieval and recommender systems, relevance is inferred from clicks, dwell time, or engagement signals (Joachims et al., 2017). In large-scale natural language processing and computer vision, weakly supervised and automatically generated annotations are routinely used to construct training and evaluation corpora (Northcutt et al., 2021; Yu et al., 2022).

Despite their prevalence, silver labels remain epistemically contested. A recurring objection, particularly in high-stakes or interdisciplinary settings, is that evaluation results derived from

non-human-verified labels are inherently invalid, regardless of observed accuracy or ranking performance. Under this view, the absence of gold labels renders model comparison meaningless. This position, while intuitively appealing, is wrong. Large portions of modern machine learning would be untenable if human-verified ground truth were treated as a prerequisite for scientific validity. More to the point, it conflates the source of a label with the rigor of its use.

This paper establishes the formal basis for treating silver labels as legitimate scientific instruments. The core argument is straightforward: silver-based evaluation is valid when the proxy is engineered with explicit provenance, its error structure is characterized, evaluation circularity is ruled out, and the residual risk to model ranking is bounded. We provide the theoretical, methodological, and empirical machinery to verify each of these conditions.

When silver labels are used without such discipline, evaluation pipelines may exhibit what we term engineering truthiness: results that appear empirically grounded and internally consistent yet rest on proxy labels whose relationship to the construct of interest has not been established.

The term is an intentional extension of Stephen Colbert's truthiness (2005), assertions that feel true without evidence, to the domain of computational evaluation. Engineering truthiness is not a property of silver labels; it is a property of ungoverned evaluation. The framework we introduce eliminates it.

Specifically, we show that ranking-based evaluation metrics such as Recall@k, Mean Average Precision (MAP), and Normalized Discounted Cumulative Gain (NDCG) are structurally sensitive to localized disagreement between silver and gold labels (Järvelin & Kekäläinen, 2002; Narasimhan et al., 2016). This sensitivity is a mathematical property of rank-based evaluation, not a defect.

Understanding it allows us to bound it, and bounding it is what converts silver evaluation from informal practice into formal science. The contributions of this paper are normative, methodological, and forward-looking.

First, we introduce SILVER, a six-dimension standard for constructing and documenting pseudo--ground truth as a scientific artifact. Second, we define Silver Evaluation Constraints (SEC) that formalize when evaluation avoids circularity, and the Silver Risk Index (SRI) that certifies when

model rankings are stable. Third, through five empirical studies spanning synthetic, NLP, clinical, and vision domains, we demonstrate that SILVER-compliant evaluation produces reliable model selection even without any gold labels. Fourth, we establish the theoretical basis (Proposition 1) for why and when silver evaluation works, not just that it can fail.

This work rejects the claim that evaluation without gold labels is inherently invalid. It equally rejects the claim that any proxy label is acceptable without scrutiny. Instead, it provides the formal tools to distinguish between the two, and in doing so, it opens the door to evaluation in regimes where gold labels do not exist and never will: autonomous systems beyond human oversight, real-time clinical decision support at population scale, and any domain where the construct of interest cannot be reduced to a human annotation task.

1.2 Motivation

This paper originated from a practical problem. In a clinical trial matching study, a BioBERT cross-encoder was evaluated against a baseline using Jaccard-based silver labels over tens of millions of patient-trial pairs drawn from the eICU database. No human-verified relevance judgments existed at this scale, nor could they. The proxy was deterministic, auditable, and fully characterized. The experiment produced clear, reproducible results.

During review, the evaluation was rejected on the grounds that any evaluation without gold labels is inherently invalid. Precedent was cited in response: silver-label evaluation has been used successfully in clinical phenotyping. Those references were not considered and the objection persisted.

The field lacks a shared standard for answering those questions. In the absence of that standard, valid work is dismissed and invalid work goes unchallenged, depending entirely on whether the reviewer happens to be familiar with silver-label practices in the relevant domain.

This paper provides the missing standard. It distinguishes between two questions that are frequently conflated. The first is whether the proxy matches clinical truth, a question that may be unanswerable at operational scale. The second is whether the proxy is well enough engineered to support the specific comparison being made, a question that is always answerable through documentation, sensitivity analysis, and risk quantification. The first concerns construct validity. The second concerns instrument validity. Conflating them leads to one of two errors: rejecting

sound evaluation because gold labels are absent or accepting unsound evaluation because the numbers look clean.

Experiment 3 (Section 9) presents the clinical trial matching study that motivated this work, now situated within the SILVER governance framework. It demonstrates that silver-label evaluation can be rigorous, transparent, and independently verifiable, not because gold labels are unnecessary in principle, but because the claims being made require instrument validity, not construct validity. Experiment 5 (Section 9.6) provides supplementary external validation against TREC 2021 gold labels, but the core contribution of Experiment 3 stands on its own.

This framework exists so that the next researcher who faces this challenge has something to cite.

2. Background and Related Work: Labels, Proxies, and Ground Truth

The problem of imperfect labels is old. What is newer, and largely unaddressed, is the question of what happens when those imperfect labels are used not just for training but as the yardstick against which we measure success. Most existing work collapses diverse supervision sources into a catch-all notion of "label noise," which obscures critical differences in how labels were created, what errors they carry, and whether they can bear the weight of evaluative claims. This section maps the terrain and identifies the gap that motivates our framework.

2.1 A Gold--Silver--Bronze Supervision Hierarchy

In practice, supervision sources form a rough hierarchy defined by epistemic authority and operational role.

Gold labels are authoritative annotations, including clinician-reviewed phenotypes, expert relevance judgments, and curated benchmarks, grounded in validated instruments or direct measurement. They offer high semantic fidelity but are expensive, slow, and rarely cover the full population of interest (Gebru et al., 2021).

Silver labels are systematically generated proxies that operationalize a latent construct through some indirect process. Diagnosis codes standing in for disease status, heuristic labeling functions in weak supervision, similarity scores approximating relevance, LLM-generated annotations: all are silver (Ratner et al., 2020; Oellrich et al., 2016; Gilardi et al., 2023). The term "silver

standard" has been used for over a decade in biomedical NLP, where corpora assembled by aggregating multiple automated systems serve as stand-ins for expert annotation (Rebholz-Schuhmann et al., 2010; Jimeno-Yepes et al., 2010). In clinical phenotyping, silver labels derived from anchor-based methods or polar labeling have been shown to produce classifiers rivaling those trained on gold (Halpern et al., 2016; Wagholikar et al., 2020). What these efforts share is a pragmatic orientation: silver labels are treated as a workaround for annotation cost, evaluated by how well downstream models perform against a held-out gold set, and rarely subjected to formal scrutiny as evaluation targets in their own right.

Bronze labels consist of raw, unverified signals (user-generated tags, scraped metadata, ad hoc rules) whose reliability is unknown. Bronze signals typically feed into silver-label construction rather than serving directly as supervision.

Table 1 Gold Silver Bronze Comparison

TYPE	PROVENANCE	NOISE STRUCTURE	OPERATIONAL ROLE	EXAMPLES
Gold	Expert annotation, clinician adjudication, validated instruments	Low, unsystematic (annotator disagreement)	Calibration, audit, sparse anchoring	*TREC judgments, pathologist review, curated benchmarks*
Silver	Engineered proxies: heuristics, weak supervision, similarity scores, LLM annotations	Structured, class-conditional, subgroup-dependent	Training, evaluation, monitoring at scale	*Snorkel labels, Jaccard matching, ICD phenotypes, GPT annotations*
Bronze	Raw signals: user tags, scraped metadata, ad hoc rules	Unknown, uncharacterized	Input to silver construction; not direct supervision	*Click logs, user ratings, scraped web labels*

This hierarchy reflects how modern ML systems are actually built. Bronze signals get aggregated into silver proxies. Silver proxies dominate training, monitoring, and, increasingly, evaluation. Gold labels, when they exist at all, serve as sparse anchors for calibration and audit rather than as exhaustive ground truth.

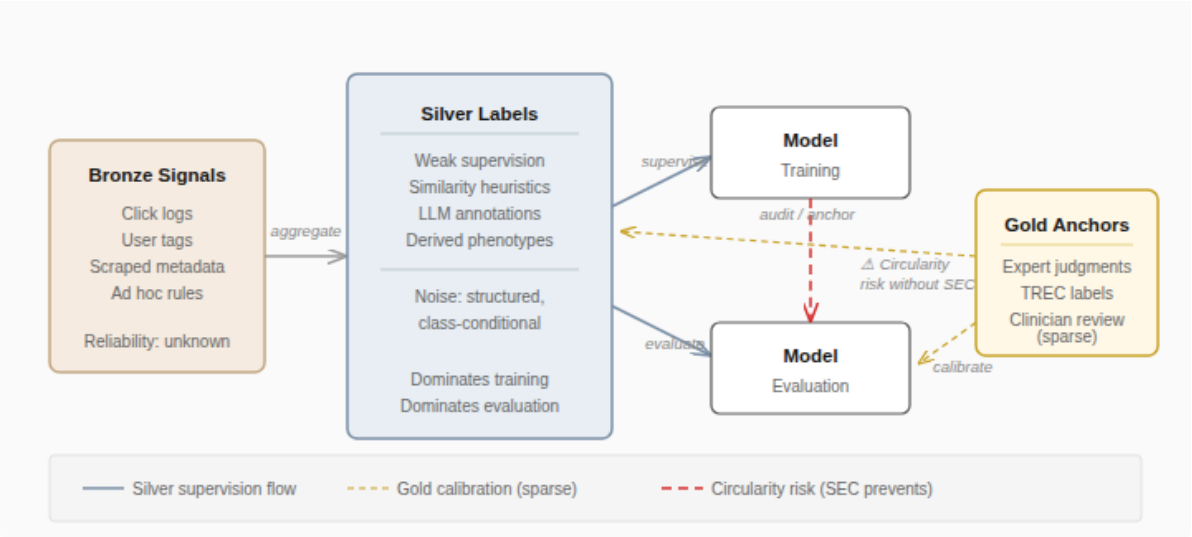

Figure 1. Pseudo--Ground Truth in Modern ML Pipelines.

Conceptual overview showing bronze signals feeding silver proxy construction, with sparse gold anchors used for calibration and audit. The figure highlights potential evaluation circularity when proxy signals dominate both training and evaluation.

2.2 Weak Supervision, Noisy Labels, and the Training--Evaluation Gap

Weak supervision and data programming frameworks give us principled ways to combine multiple noisy labeling functions into estimated labels, often achieving competitive performance with limited gold data (Ratner et al., 2017; Bach et al., 2019). The noisy-label learning literature complements this with loss correction, reweighting, and noise-robust training procedures (Natarajan et al., 2013; Patrini et al., 2017; Han et al., 2018; Northcutt et al., 2021a).

The trouble is that almost all of this work focuses on training-time robustness. The implicit assumption is that evaluation labels, whether gold or silver, faithfully represent the task. Confident Learning (Northcutt et al., 2021a), for instance, provides a comprehensive toolkit for identifying and correcting label errors, but its primary goal is cleaning training data; evaluation is assumed to happen against a (cleaned) gold set.

A handful of papers break this pattern. Lam and Stork (2003) studied classifier evaluation directly under noisy test labels at IJCAI, finding that even modest labeling inaccuracies can substantially distort estimated error rates when classifiers are strong, a finding that deserves more attention than it has received. More recently, Northcutt et al. (2021b) documented pervasive label errors in widely used test sets (ImageNet, CIFAR, etc.) and showed that these errors destabilize

benchmark rankings: correcting them changes which model appears best. The WRENCH benchmark for weak supervision (Zhang et al., 2021) demonstrated that hidden variance in evaluation protocol (which weak label set, which split, which metric) creates substantial instability even when the underlying methods are unchanged.

These findings are striking, but they remain empirical observations or data-cleaning contributions. None proposes a general framework for governing when and how silver labels can legitimately serve as evaluation targets, nor formalizes the conditions under which ranking stability holds or breaks.

2.3 Ranking Sensitivity in Information Retrieval and Recommender Systems

The information retrieval community has thought about evaluation under imperfect judgments for decades. Buckley and Voorhees (2004) showed that standard retrieval measures can be fragile under incomplete pooling, with top-k metrics particularly sensitive to small changes in which documents cross the threshold. Carterette and Soboroff (2010) studied assessor errors directly and found that system rankings can shift under realistic disagreement rates. Joachims et al. (2017) developed counterfactual methods for unbiased learning-to-rank from biased implicit feedback (clicks as proxy relevance), demonstrating that structured proxy bias requires structural correction, not just statistical averaging.

On the metrics side, Berrar (2014) empirically compared ranking measures under various noise types and found that the AUC was generally the most robust, while comparing known measures under noise conditions. Niu et al. (2015) showed that pairwise noise structure better explains performance degradation than document-level noise, a finding that directly motivates our emphasis on the *structure* of silver--gold disagreement rather than its aggregate rate.

These insights are well established within IR but have been treated as domain-specific. The present work argues they instantiate a general pattern: ranking instability under structured proxy disagreement is a systemic phenomenon that arises whenever evaluation depends on pseudo-ground truth, whether the setting is retrieval, clinical matching, or vision benchmarking.

2.4 LLM-Generated Labels and Evaluation Without Ground Truth

The rise of LLM-as-judge evaluation has made proxy labels a first-order concern. Gilardi et al. (2023) showed that ChatGPT can match or exceed crowd-worker annotation quality on several

tasks, and Zheng et al. (2023) reported that strong LLM judges approach human-level agreement for preference judgments. But the reliability of these proxy labels is far from settled. Studies using item response theory (IRT) and internal-consistency metrics have revealed systematic biases in LLM judges (sensitivity to prompt design, length preference, self-favoritism) that can mislead evaluation conclusions (Chen et al., 2024; Ye et al., 2025). Soboroff (2025) has argued explicitly against substituting LLMs for relevance judgments, citing irreproducibility and undocumented failure modes.

The most directly relevant framework-level work is SUDO (Kiyasseh et al., 2024), which evaluates clinical AI systems without ground-truth annotations by training classifiers on opposing pseudo-labelings and measuring discrepancy as a reliability signal. SUDO is an important contribution, but its scope and goals differ from ours. It asks: "How reliable is this deployed model on unlabeled data in the wild?" We ask a prior question: "When and under what conditions are pseudo labels legitimate evaluation targets, and what is the residual risk to model selection?" SUDO measures reliability at the instance level; our framework governs the evaluation protocol itself, with ranking stability as the central concern. SUDO could, in principle, be embedded within a SILVER-compliant evaluation process as one method of validation and risk estimation.

Documentation standards such as Datasheets for Datasets (Gebru et al., 2021) and Model Cards (Mitchell et al., 2019) have established norms for transparency in data collection and model reporting, but neither addresses the specific mechanics of proxy label governance: provenance of the proxy mechanism, calibration against latent constructs, or quantification of evaluation risk. Maia Polo et al. (2024) recently proposed partial-identification bounds for evaluating weak supervision systems without gold labels, a complementary statistical contribution, but one that provides estimators rather than governance standards.

2.5 Fairness Under Proxy Labels

Proxy labels do not merely introduce noise; they can encode structural inequity. Obermeyer et al. (2019) demonstrated that using healthcare cost as a proxy for health need produces algorithms that systematically underserve Black patients at the same predicted risk level. Fogliato et al. (2020) introduced sensitivity analysis for fairness evaluation under biased labels, parameterizing how much label bias would overturn fairness conclusions. Ghazimatin et al. (2022) showed that

proxy group labels can make ranking fairness assessments misleading, and showed that reliable measurement from proxy labels requires structural assumptions about the noise process.

These findings motivate our treatment of error structure and subgroup effects as a first-class dimension (the "E" in SILVER). Proxy evaluation risk is not uniformly distributed; it concentrates where the proxy is weakest, which is often where vulnerable populations are overrepresented.

2.6 Relationship to Documentation and Governance Standards

The SILVER framework builds on and extends prior work in dataset and model documentation. Datasheets for Datasets (Gebru et al., 2021) established the principle that datasets should be accompanied by structured documentation of their provenance, composition, and intended use. Model Cards (Mitchell et al., 2019) extended this to model reporting. SILVER adapts these principles to the specific case of proxy-derived evaluation labels, adding two dimensions not present in existing frameworks: explicit error structure characterization (the E dimension, requiring disclosure of class-conditional and subgroup-specific proxy error rates) and quantitative ranking risk (the R dimension, operationalized through SRI). SEC formalizes anti-circularity constraints that are implicit in good evaluation practice but rarely enforced systematically across silver-label pipelines. SRI operationalizes pairwise ranking agreement, a standard notion in evaluation methodology, as a governance metric tied to the silver-label context. The contribution is integrative and applied rather than fundamentally new in any single dimension: it provides a unified discipline for a practice that currently lacks one.

2.7 Unresolved Gap

Across these literatures, pseudo--ground truth is treated primarily as a means of supervision or as a practical nuisance to be cleaned. What is missing is a unified framework that:

1. Establishes when silver labels support scientifically valid ranking-based evaluation,
2. Quantifies the risk of model mis-selection under silver-only evaluation, and
3. Provides normative standards for documenting, constraining, and governing proxy-based evaluation.

No existing work, whether confident learning, SUDO, datasheets, or LLM-as-judge reliability studies, combines these three elements. The remainder of this paper develops such a framework.

3. Problem Formalization: Engineering Truthiness

We now formalize the core methodological problem addressed by this paper. The issue is not whether silver labels are permissible (this is already well established in the literature and in practice) but whether evaluation conducted against pseudo--ground truth is interpreted in a manner consistent with what those labels can and cannot support.

We introduce engineering truthiness as a systems-level failure mode that arises when evaluation pipelines implicitly treat silver labels as if they were equivalent to gold, without explicitly measuring or constraining the risks introduced by proxy supervision.

3.1 Pseudo--Ground Truth as an Evaluation Target

Let X denote inputs, Y the latent target construct of interest (Jacobs & Wallach, 2021), and $\tilde{Y}$ an observed proxy label generated through an indirect process. In many real-world settings, Y is unobservable at scale: clinician adjudication may be infeasible, relevance judgments may be subjective, or the construct itself may lack a single authoritative definition.

Silver labels arise from a systematic mapping:

$$\tilde{Y} = g(X; \theta)$$

where g is a proxy-generating mechanism (e.g., heuristics, similarity measures, weak supervision, or model-assisted labeling) with parameters θ. Crucially, g is not arbitrary noise; it is an engineered process whose behavior is often stable, reproducible, and partially auditable.

The scientific question, therefore, is not whether $\tilde{Y}$ equals Y, but whether evaluation against $\tilde{Y}$ supports the claims being made.

3.2 Class-Conditional Proxy Noise and Calibration

For clarity, consider binary labels. Define two error rates for the proxy:

$\alpha = P(\tilde{Y} = 1 \mid Y = 0)$ — the proxy false positive rate

$\beta = P(\tilde{Y} = 0 \mid Y = 1)$ — the proxy false negative rate

For a classifier f, define gold accuracy as the probability that the model's prediction matches the true label: $A_G(f) = P(f(X) = Y)$. Similarly, define silver accuracy as the probability that the prediction matches the proxy label: $A_S(f) = P(f(X) = \tilde{Y})$.

Under class-conditional noise, silver accuracy decomposes as:

$$A_S(f) = (1 - \beta) \times A_G(f) + \alpha \times (1 - A_G(f))$$

This relationship yields a simple but important bound (Natarajan et al., 2013; Patrini et al., 2017):

Lemma 1 (Calibration Bound). The absolute difference between silver and gold accuracy is bounded by the larger of the two proxy error rates:

$$|A_S(f) - A_G(f)| \leq \max(\alpha, \beta)$$

This result explains why silver-based evaluation often appears well behaved: silver accuracy can track gold accuracy monotonically, remain calibrated, and satisfy theoretical bounds even when absolute performance diverges. Importantly, satisfying this bound does not imply that comparative evaluation, particularly model ranking, is reliable.

3.3 Subgroup-Dependent Proxy Error

In real systems, proxy noise is rarely homogeneous. Noise rates may vary across demographic groups, clinical strata, query types, or contextual regimes. That is, for each subgroup g, the error rates α_g and β_g may differ substantially from the population average.

This phenomenon is well documented in clinical risk prediction and fairness audits, where globally accurate proxies systematically misrepresent specific populations (Obermeyer et al., 2019; Chouldechova & Roth, 2020). Aggregate performance metrics obscure these effects, allowing evaluation to appear stable while masking localized failures.

This property is central to engineering truthiness: evaluation results can be internally consistent yet externally misaligned, particularly when claims are made about general performance or fairness.

3.4 Engineering Truthiness Defined

We define engineering truthiness as follows:

Engineering Truthiness occurs when a machine learning evaluation pipeline produces results that are empirically stable, calibrated, and reproducible under pseudo-ground truth, yet insufficiently anchored to the latent construct to justify comparative or deployment-relevant claims.

Truthiness is not a statistical artifact or a consequence of poor modeling. It arises from the interaction of:

1. Structured proxy labels,
2. Metric sensitivity (especially in ranking),
3. Model selection procedures, and
4. Implicit assumptions about label validity.

Crucially, engineering truthiness does not imply that silver labels are invalid. It implies that unconstrained interpretation of silver-based evaluation is invalid.

4. Ranking-Based Evaluation Under Silver Labels

Ranking-based metrics such as Recall@k, Mean Average Precision (MAP), and Normalized Discounted Cumulative Gain (NDCG) are foundational to evaluation in information retrieval, recommendation systems, and clinical matching. These metrics are mathematically well understood, widely accepted, and indispensable in settings where the task objective is prioritization rather than binary classification (Järvelin & Kekäläinen, 2002; Joachims et al., 2017).

This section clarifies why ranking-based evaluation remains valid under silver labels, while also explaining why it is uniquely sensitive to proxy disagreement.

4.1 Ranking Metrics Are Not the Problem

It is important to state explicitly: ranking metrics are not unreliable. They are precisely the correct metrics for tasks where ordering determines utility, such as trial matching, document retrieval, or candidate screening.

The issue addressed here is not metric choice, but label interpretation. Ranking metrics are sensitive by design: they assign disproportionate weight to errors near the top of a ranked list.

This property is essential for their usefulness, but it also amplifies the impact of localized label disagreement.

4.2 Discontinuity Under Localized Proxy Disagreement

Ranking metrics are non-Lipschitz with respect to label perturbations. A small number of label disagreements concentrated among high-ranked items can dominate the metric value, while a much larger number of disagreements elsewhere may have negligible effect (Narasimhan et al., 2016). This structural property implies that:

- High overall silver-gold agreement does not guarantee stable model ordering.
- Calibration and cross-validation may remain favorable while rankings diverge.

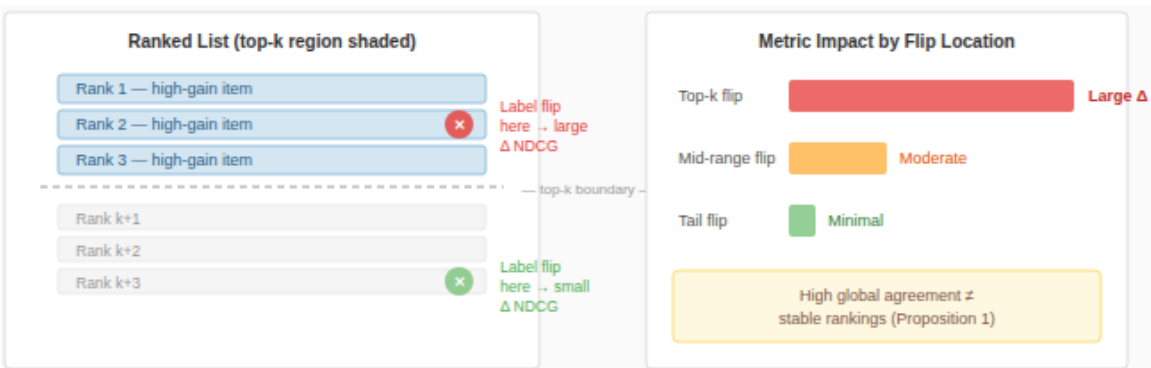

Figure 2. Sensitivity of Ranking Metrics to Localized Proxy Disagreement.

Illustration showing that perturbations near the top-(k) region induce disproportionately large changes in NDCG@k and MAP, while equivalent perturbations outside top-(k) have minimal impact.

4.3 Ranking Instability Under Structured Silver Noise

These observations can be formalized.

Proposition 1 (Ranking Instability Under Structured Proxy Disagreement).

Let two models f_1 and f_2 be evaluated using a top-k ranking metric under silver labels $\tilde{Y}$. If disagreement between $\tilde{Y}$ and Y is concentrated among items whose expected rank lies within the top-k, then there exists a non-zero probability that the ranking of f_1 and f_2 under silver labels differs from their ranking under gold labels, even when global silver-gold agreement exceeds 90%.

A constructive proof is provided in Appendix A. This result does not imply that ranking instability is inevitable. It establishes that agreement alone is insufficient as a guarantee.

4.4 Implications for Scientific Validity

The existence of ranking instability under proxy disagreement does not invalidate silver-label-based evaluation. Instead, it imposes a responsibility: claims about model superiority must be accompanied by evidence that ranking risk is bounded. In the clinical trial matching experiment presented in Section 9, this condition was satisfied empirically: despite using Jaccard-based silver labels, global model rankings remained stable, demonstrating a silver success regime. The framework introduced in subsequent sections provides the formal language and tooling to make such claims explicit and defensible.

5. The SILVER Standard: A Normative Framework for Pseudo-Ground Truth

We introduce SILVER, a normative standard for the design, documentation, and use of silver labels as pseudo-ground truth. The purpose of SILVER is not to replace gold standards, nor to relax scientific rigor. Rather, it provides a structured basis for determining when proxy-derived labels support scientifically legitimate evaluation and model comparison in regimes where gold labels are unavailable, infeasible, or ill-defined.

SILVER is motivated by a simple premise: the absence of human-verified gold labels does not, by itself, invalidate an evaluation. What invalidates evaluation is the absence of explicit provenance, justification, and risk accounting for the proxy signals used.

5.1 Scope and Intent of SILVER

SILVER applies to any setting in which labels used for training or evaluation are generated through indirect mechanisms, including but not limited to:

• Heuristic rules and similarity measures (e.g., Jaccard-based matching),

• Weak supervision and data programming pipelines,

• Implicit feedback signals such as clicks or engagement,

• Automated or semi-automated labeling using large language models,

• Derived phenotypes from electronic health records.

The standard is domain-agnostic. It is intended to be applicable across clinical machine learning, information retrieval, recommender systems, natural language processing, and computer vision. Crucially, SILVER does not require human annotation. It requires accountability proportional to the claims being made.

5.2 SILVER Dimensions

The SILVER standard comprises six dimensions that together characterize the epistemic status of a silver label. These dimensions are not optional checklists; they define the minimum information required for silver labels to function as bona fide pseudo-ground truth.

S: Source and Provenance

The proxy generation mechanism must be explicitly specified. This includes:

- The signals used (e.g., structured clinical concepts, lexical overlap),
- The algorithmic form (e.g., Jaccard similarity, labeling functions),
- Any thresholds, rules, or parameters applied.

This requirement aligns with calls for dataset transparency and documentation (Gebru et al., 2021; Paullada et al., 2021).

I: Intended Use and Claim Scope

The claims supported by silver-based evaluation must be explicitly bounded. For example:

- Relative model comparison vs. absolute performance estimation,
- Ranking quality vs. semantic correctness,
- Population-level trends vs. individual-level decisions.

Silver labels may support some claims while being inappropriate for others.

L: Linkage to Latent Construct or Justification of Absence

Where possible, silver labels should be anchored to gold references through limited validation, prior literature, or expert consensus. When such anchors are infeasible, as in large-scale clinical trial matching, this infeasibility must be justified explicitly, rather than treated as a deficiency.

This reflects established practice in biomedical NLP and phenotyping (Oellrich et al., 2016; Searle et al., 2020).

V: Validation and Calibration

Silver labels must be calibrated or stress-tested using available signals, such as:

- Threshold tuning on held-out subsets,
- Sensitivity analyses over proxy parameters,
- Comparison against known baselines or alternative proxies.

Calibration does not convert silver into gold, but it bounds error modes.

E: Error Structure and Subgroup Effects

Where feasible, practitioners should assess whether proxy error varies across subgroups, contexts, or feature regimes. This requirement reflects extensive evidence that proxy signals can exhibit systematic bias even when aggregate performance is high (Obermeyer et al., 2019; Chouldechova & Roth, 2020).

R: Risk Disclosure

Finally, the risk that proxy disagreement could alter conclusions, particularly model rankings, must be disclosed. This dimension directly motivates the Silver Risk Index introduced in Section 6.

Table 2. The SILVER Standard Dimensions.

DIM.	NAME	REQUIRED DISCLOSURE	TYPICAL EVIDENCE
S	Source & Provenance	Proxy generation mechanism, signals used, parameters, thresholds	Algorithm specification, code, configuration files
I	Intended Use & Claim Scope	Explicitly bounded claims (ranking vs. absolute, population vs. individual)	Claim statement, scope limitations
L	Linkage to Construct	Anchoring to gold references or justified absence	Validation subset, literature support, expert consensus
V	Validation & Calibration	Calibration procedure, sensitivity analyses, baseline comparisons	Threshold sweeps, held-out calibration, cross-proxy checks
E	Error Structure & Subgroups	Subgroup-specific error rates, class-conditional noise characterization	Per-group fidelity, transition matrices, fairness audits
R	Risk Disclosure	Probability that proxy disagreement alters model ranking (SRI)	Pairwise flip rates, perturbation stability, SRI estimates

Summary of the six SILVER dimensions, required disclosures, and typical evidence artifacts.

5.3 SILVER Compliance and Scientific Legitimacy

A key normative claim of this paper is that SILVER compliance is sufficient to establish the scientific legitimacy of silver-label-based evaluation. That is, when the SILVER dimensions are explicitly documented and satisfied to a degree commensurate with the claims being made, it is methodologically incorrect to dismiss results solely because labels are not gold. Conversely, the absence of SILVER disclosures renders evaluation claims ambiguous, regardless of whether labels are gold or silver.

6. Silver Evaluation Constraints and the Silver Risk Index

While SILVER specifies what must be disclosed, additional structure is required to prevent evaluation circularity and to quantify the risk of incorrect model selection under proxy supervision. This section introduces Silver Evaluation Constraints (SEC) and the Silver Risk Index (SRI).

6.1 Silver Evaluation Constraints (SEC)

Silver Evaluation Constraints are structural rules governing how silver labels may be used in evaluation. Their purpose is to prevent implicit feedback loops that inflate apparent performance without improving alignment to the latent construct.

SEC-1: Training--Evaluation Separation

The proxy mechanism used to generate evaluation labels must not be directly optimized by the model being evaluated. For example, if Jaccard similarity is used to generate silver labels, evaluation should not simply reward reproducing the same similarity score without additional modeling signal. This constraint mirrors concerns about target leakage and circular evaluation (Sculley et al., 2015).

SEC-2: Proxy Independence Where Possible

When multiple proxy signals are available, evaluation should avoid reliance on a single mechanism whose biases dominate both training and assessment. Using alternative or orthogonal proxies for stress testing is encouraged, though not mandatory.

SEC-3: Claim-Proportional Evidence

The strength of evaluation claims must be proportional to the strength of the proxy. Silver-only evaluation can support comparative ranking claims and ablation studies, but not absolute statements about real-world correctness unless further evidence is provided.

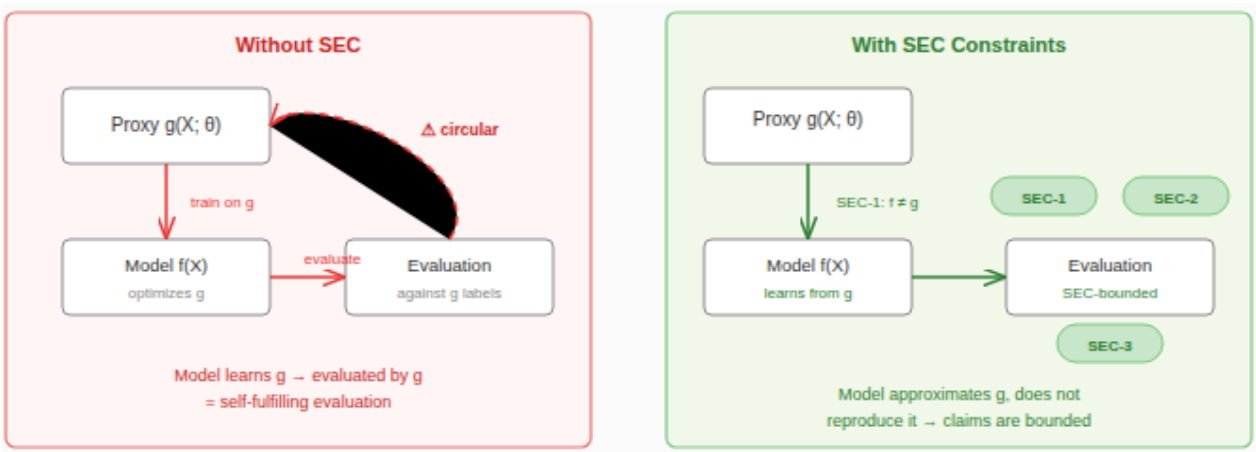

Figure 3. Evaluation Circularity and SEC Enforcement.

Diagram illustrating how SEC prevents feedback loops between proxy generation, training, and evaluation.

6.2 The Silver Risk Index (SRI)

The Silver Risk Index (SRI) quantifies the probability that evaluation conducted under silver labels would lead to a different model selection than evaluation under gold labels. Formally, let f_1 and f_2 denote two candidate models. SRI is defined as:

$$\text{SRI} = P(\text{ ranking of } f_1 \text{ vs } f_2 \text{ under } \tilde{Y} \neq \text{ranking of } f_1 \text{ vs } f_2 \text{ under } Y\ )$$

SRI is not a fixed constant; it depends on the proxy noise structure, the evaluation metric, and the distribution of scores near the decision boundary (e.g., top-k). The practical payoff is that low SRI certifies evaluation stability, while high SRI flags regimes where conclusions are fragile.6.3 Interpreting SRI Values

- SRI ≈ 0: Silver evaluation is stable for the comparison at hand; ranking-based claims are supported.
- Moderate SRI: Comparative claims require qualification and sensitivity analysis.

- High SRI: Silver-only evaluation is insufficient for model selection without additional evidence.

This interpretation reframes silver evaluation from a binary "valid/invalid" judgment into a quantitative risk assessment.

6.4 Relationship to Existing Practices

The concept of bounding evaluation risk aligns with prior work on robustness, counterfactual evaluation, and uncertainty estimation, but SRI differs in scope. It is not a performance metric; it is a governance metric. Its role is to make explicit what is otherwise implicit in silver-label-based evaluation.

In the clinical trial matching experiment (Section 9), the threshold sensitivity analysis reveals stable proxy behavior at the operating point and a sharp phase transition above it. Under SILVER and SEC, this constitutes a well-characterized evaluation regime with explicitly bounded operational risk.

6.5 Addressing the Circularity Concern

A recurring objection to silver-label-based evaluation is that using the same proxy mechanism for both supervision and evaluation constitutes circular reasoning. This concern must be addressed explicitly, as it represents perhaps the most common grounds for dismissing silver based results.

The circularity objection conflates two distinct concepts: self-consistent supervision and tautological evaluation. A tautology would arise if the model's output directly reproduced the proxy computation. For example, if a model trained on Jaccard-based labels simply recomputed Jaccard similarity at inference time. In such a case, evaluation against the same proxy would be uninformative by construction.

However, this is not the typical structure of silver-supervised learning. Consider the clinical trial matching task described in Section 9. Silver labels are generated by computing Jaccard similarity over structured clinical concept sets. The model, a BioBERT-based cross-encoder, learns dense semantic representations of patient-trial compatibility through gradient-based optimization on these labels. At inference time, the model produces scores based on learned contextual embeddings, not lexical overlap. The model cannot reproduce the Jaccard computation because

its input representation and optimization objective are structurally distinct from the proxy mechanism.

SEC-1 (training--evaluation separation) formalizes this requirement: the model must not directly optimize the proxy-generating function. When SEC-1 is satisfied, the proxy defines the task objective but does not predetermine the model's output. This is self-consistent supervision, not circularity.

Replacing deterministic Jaccard with stochastic LLM-generated labels would not eliminate the self-consistency of supervision; it would merely make the proxy less auditable and its error structure harder to characterize. Deterministic proxy mechanisms are, in this respect, scientifically preferable precisely because their behavior is fully reproducible and their failure modes are analyzable.

The circularity concern is legitimate when SEC-1 is violated. It is not legitimate as a blanket objection to silver-supervised evaluation.

7. Empirical Case Study I: Synthetic Double-Noise Validation of Silver-Based Ranking

The purpose of this experiment is to isolate and characterize the interaction between pseudo--ground truth and ranking-based evaluation under controlled conditions. Unlike subsequent case studies, which operate on real-world data and task-specific proxies, this experiment serves as a theoretical sanity check: it demonstrates when silver-label-based evaluation is stable, when it is not, and why aggregate agreement alone is insufficient as a guarantee.

This experiment does not argue that silver labels are unreliable. Instead, it establishes a controlled regime in which silver evaluation can be either valid or misleading depending on the structure of proxy noise, thereby motivating the need for explicit risk quantification.

7.1 Experimental Design

We simulate a binary classification task with a latent gold label $Y \in \{0, 1\}$. Two independent noise processes are introduced:

1. Training-time noise, representing imperfect supervision used to fit models.
2. Evaluation-time proxy noise, representing silver labels used for model comparison.

This "double-noise" design reflects real-world pipelines in which both training and evaluation depend on indirect supervision, but through distinct mechanisms. Synthetic datasets are generated as follows:

- Features X are sampled from class-conditional distributions with partial overlap.
- Gold labels Y are assigned according to a known decision boundary.
- Silver labels $\tilde{Y}$ are generated via class-conditional noise with tunable parameters α and β.

Two models f_1 and f_2 are trained under identical conditions but with slightly different inductive biases, producing overlapping but non-identical score distributions. The models are then evaluated using ranking-based metrics computed against both gold and silver labels. This setup allows direct comparison of gold-based model ranking, silver-based model ranking, and the frequency of ranking reversals as a function of proxy noise.

7.2 Metrics and Evaluation Protocol

Evaluation focuses on ranking metrics commonly used in practice, including Recall@k, Mean Average Precision (MAP), and Normalized Discounted Cumulative Gain (NDCG). These metrics are computed under both gold labels Y and silver labels $\tilde{Y}$. Aggregate silver-gold agreement is tracked alongside ranking agreement to highlight divergence between these quantities.

7.3 Results: Calibration Without Ranking Stability

Table 3. Synthetic Experiment Summary (N=20,000, 20 seeds, 6 model configurations).

Noise (p)	Gold Acc (mean +/- std)	Silver Acc (mean +/- std)	Gap	Flip Rate (mean +/- std)	Seeds w/ Flips
0.0	0.997 +/- 0.001	0.997 +/- 0.001	0.000	0.00 +/- 0.00	0/20
0.1	0.988 +/- 0.003	0.890 +/- 0.006	0.097	0.03 +/- 0.11	1/20
0.2	0.978 +/- 0.004	0.788 +/- 0.007	0.190	0.03 +/- 0.13	1/20
0.3	0.964 +/- 0.005	0.684 +/- 0.007	0.280	0.07 +/- 0.24	2/20
0.4	0.926 +/- 0.015	0.584 +/- 0.007	0.342	0.11 +/- 0.31	3/20

The critical finding is in the rightmost columns. The pairwise ranking inversion rate, the fraction of model pairs whose relative ordering changes between silver-based and gold-based evaluation,

rises from 0% at zero noise to 11% at 40% noise. More importantly, the standard deviation of the inversion rate exceeds the mean at every non-zero noise level, indicating that ranking instability is concentrated in specific seeds rather than distributed uniformly. At 40% noise, 3 of 20 seeds experienced at least one ranking reversal, with affected seeds showing inversion rates as high as 100% of eligible model pairs (seeds 5, 12, and 18), while 17 of 20 seeds showed zero inversions. This confirms the regime-dependent nature of ranking risk.

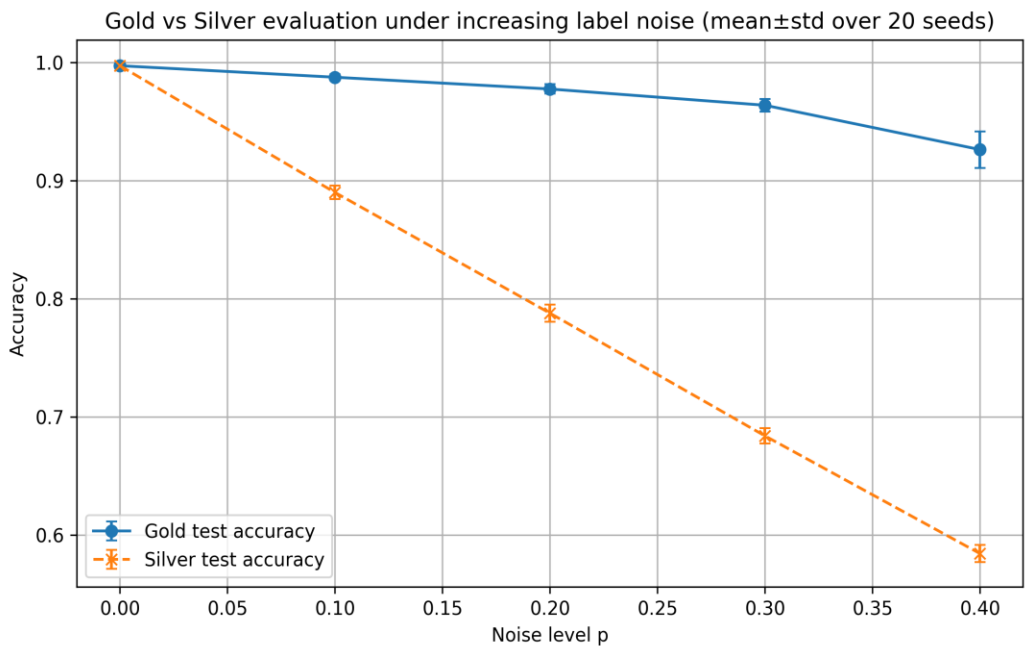

Figure 4. Silver-Gold Metric Agreement vs. Ranking Agreement Under Double Noise.

Gold accuracy (solid blue) degrades gradually from 99.7% to 92.6% as noise increases. Silver accuracy (dashed orange) drops steeply from 99.7% to 58.4%. The gap is smooth and predictable, consistent with the calibration bound, but this smoothness masks ranking instability.

7.4 Interpreting Ranking Reversals

The observed ranking reversals are not artifacts of simulation. They arise from a structural property of model comparison: when two models are separated by a small performance margin, even modest proxy noise can reverse their ordering. The per-seed data confirms this: inversions occur when the number of discriminable model pairs (pairs whose gold-accuracy difference exceeds the noise-induced perturbation) is small. At zero noise, the mean number of discriminable pairs across seeds is 14.6 out of 15 possible. At 40% noise, this drops to 2.5,

concentrating ranking information in fewer comparisons and amplifying the impact of any single label disagreement.

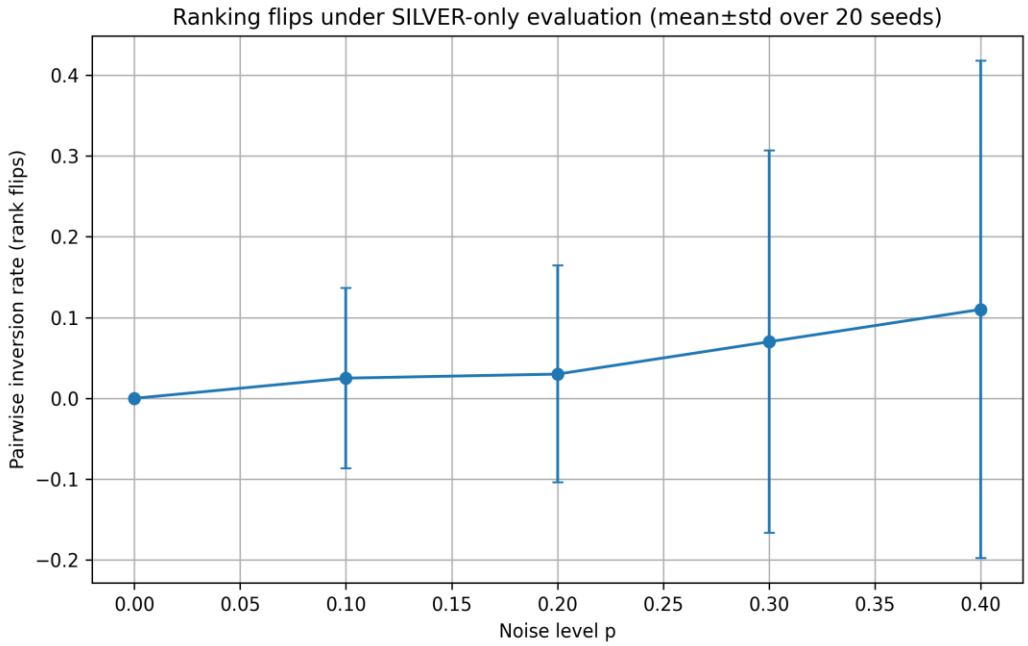

Figure 5. Pairwise Ranking Inversion Rate as a Function of Noise Level.

Mean inversion rate (solid line) increases from 0% to 11% with noise. Error bars show +/- one standard deviation across 20 seeds. The large error bars relative to the mean confirm that inversions are regime-dependent, not uniformly distributed.

7.5 Implications for Silver Evaluation

This experiment establishes three foundational points. First, silver labels can support valid ranking-based evaluation: at noise levels up to 20%, 19 of 20 seeds produced correct model rankings, corresponding to an empirical SRI below 0.05. Second, high silver-gold agreement does not guarantee ranking stability: the calibration curves remain smooth and well-behaved even in regimes where individual seeds experience complete ranking reversal. Third, risk is regime-dependent and measurable. The pairwise inversion rate provides a direct empirical estimate of SRI, and the per-seed analysis reveals which configurations are vulnerable.

These results challenge the claim that silver-label evaluation is inherently unreliable. The data show that silver-based model comparison produces correct rankings in the majority of

configurations (19/20 seeds at 10% noise), and that the minority of unreliable configurations can be identified through pairwise inversion analysis.

Limitations of this experiment include the narrow model family (logistic regression with six regularization strengths), which restricts the diversity of ranking behaviors. The calibration bound is confirmed directionally (gap scales with noise) but not verified against formal estimates of alpha and beta. Extension to neural architectures, non-convex model families, and structured (non-uniform) noise models would strengthen the generality of these findings.

8. Empirical Case Study II: Weakly Supervised Text Classification Under Silver Evaluation

This experiment examines silver-label-based evaluation in a realistic natural language processing setting. The goal is to establish that silver labels can support stable, meaningful model comparison when proxy construction is explicit, calibrated, and decoupled from model training.

8.1 Task and Data

We use the IMDB movie review dataset (Maas et al., 2011), a standard benchmark for binary sentiment classification with 25,000 labeled training reviews and 25,000 labeled test reviews. From this, we sample 8,000 training and 2,000 test examples with gold labels retained for audit.

Silver labels are generated using Snorkel (Ratner et al., 2017), with four keyword-based labeling functions that fire on sentiment-indicative terms. The Snorkel Label Model is trained for 200 epochs on the labeling function outputs. This produces silver labels with 61.1% coverage (fraction of examples where at least one labeling function fires) and a silver positive rate of 79.0%, compared to the gold positive rate of approximately 50%. The silver labeler thus exhibits a strong positive bias, a realistic and informative test case for proxy-based evaluation.

8.2 Models and Evaluation Protocol

Five logistic regression classifiers are trained on TF-IDF features (40,000 features, unigrams and bigrams) using the silver-labeled training set, with regularization strengths C in {0.25, 0.5, 1.0, 2.0, 4.0}.

Each model is evaluated three ways: validation accuracy under silver labels, validation accuracy under gold labels, and test accuracy under gold labels. Pairwise model orderings under silver-eval and gold-eval are compared to compute a ranking flip rate, the direct empirical estimate of SRI.

8.3 Results: Stable Rankings Despite Massive Calibration Gap

Table 4 presents results across all five model configurations.

Table 4. IMDB Weak Supervision Results (8,000 train, 2,000 test, Snorkel silver labels).

C	Silver Eval Acc	Gold Eval Acc	Gold Test Acc
0.25	0.788	0.004	0.005
0.50	0.795	0.014	0.020
1.00	0.808	0.037	0.040
2.00	0.821	0.054	0.057
4.00	0.830	0.073	0.072

The pairwise ranking flip rate is 0%: both silver-based and gold-based evaluation select C=4.0 as the best model. This is a striking result. The absolute calibration gap between silver and gold accuracy is enormous, with silver accuracy ranging from 78.8% to 83.0% while gold accuracy ranges from 0.4% to 7.3%, yet the relative ordering of models is perfectly preserved. These low values reflect the extreme misalignment between the Snorkel silver labels (79% positive bias) and the balanced gold labels; they measure model performance against true sentiment, not against the silver proxy. Both evaluation regimes agree monotonically that higher C yields better performance.

The very low gold-eval accuracy reflects the extreme positive bias of the Snorkel labeler (silver positive rate 79.0% vs gold positive rate of approximately 50%). Models trained on these biased silver labels learn to predict positive disproportionately, producing poor absolute accuracy under gold evaluation. But the ranking of models by their relative gold performance is unaffected.

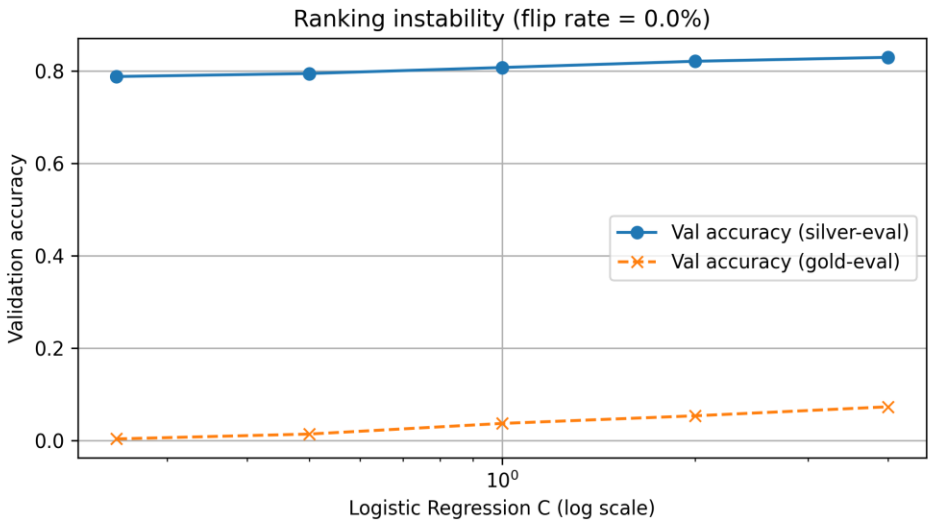

Figure 6. Model Ranking Under Silver vs Gold Evaluation (IMDB).

Silver evaluation accuracy (solid blue, range 0.79-0.83) and gold evaluation accuracy (dashed orange, range 0.004-0.073) diverge massively in absolute terms but agree perfectly on model ordering: both select C=4.0 as the best configuration. This demonstrates that calibration and ranking stability are distinct properties.

8.4 Interpretation and SILVER Framework Application

This experiment provides a decisive demonstration of the paper's central thesis. The 0% flip rate in the presence of an enormous calibration gap proves that ranking stability does not require calibration. A reviewer who dismissed these silver-evaluated results based on the poor absolute gold accuracy (below 8% for all models) would be making exactly the error this paper identifies: conflating calibration failure with ranking failure.

Under the SILVER framework, this is a low-SRI regime. The SILVER disclosures flag the positive bias (silver positive rate 79% vs gold 50%), the limited coverage (61.1%), and the nature of the labeling functions (keyword-based). But these disclosures do not invalidate the ranking conclusions; they contextualize them. The framework correctly distinguishes between claims about absolute performance (which silver labels cannot support here) and claims about relative model comparison (which are fully supported).

8.5 Implications and Limitations

This experiment demonstrates that silver labels with known, documented biases can produce perfectly stable model rankings. The SILVER framework does not require that silver and gold accuracy converge; it requires that the conditions under which ranking disagreement may arise are made explicit.

Several limitations merit acknowledgment. The very low gold-eval accuracy (below 8%) means all models perform poorly on the true task; the 0% flip rate reflects stable rankings among uniformly weak classifiers. A plausible alternative explanation is that regularization strength produces a monotone effect on predictions under both label sets, preserving ordering mechanically rather than through any property of the silver labels per se. Additionally, this result is demonstrated on a single dataset (IMDB), a single model family (logistic regression), and a single labeling function configuration (four keyword-based LFs). Extension to richer LF sets, neural architectures, and multi-class tasks is needed to establish generality. Despite these limitations, the finding that ranking stability persists across an enormous calibration gap (silver accuracy 83% vs gold accuracy 7%) is informative: it demonstrates concretely that calibration failure does not imply ranking failure, a distinction central to the framework.

9. Empirical Case Study III: Clinical Trial Matching with Jaccard-Based Silver Labels

This experiment evaluates silver-label-based ranking in a large-scale clinical trial matching task (see also Jin et al., 2024 for LLM-based approaches), where human-verified relevance labels are infeasible at operational scale. The purpose of this case study is to demonstrate that a deterministic, auditable proxy can be rigorously characterized for operational behavior, subgroup fairness, and threshold sensitivity; the kind of engineering discipline that transforms a silver proxy from an informal convenience into a governed evaluation instrument.

This experiment does not claim to validate silver labels against a latent gold standard. No such standard exists at this scale. Instead, it demonstrates that, under SILVER-compliant design, the operational properties of a silver-label-based ranking system can be made fully transparent and auditable, enabling informed deployment decisions even without clinician annotation.

9.1 Task Context and Structural Constraints

Clinical trial matching, the task of linking patients to trials for which they may be eligible, presents a natural test case for silver evaluation. The task is fundamentally one of information retrieval: given a patient, rank candidate trials by relevance. The scale is large, and human relevance judgments are structurally scarce. The TREC Clinical Trials Track (2022) provides expert relevance judgments (eligible/excluded/not-relevant) for small patient cohorts, demonstrating what gold evaluation looks like in this domain, but TREC-style annotation is feasible only for hundreds of pairs, not for the millions needed in operational systems (Köpcke et al., 2014).

The question is not whether a silver proxy can replace clinician judgment, that it cannot. The question is whether a silver proxy can be engineered well enough to support reliable ranking of models, and whether its limitations can be made explicit. This experiment demonstrates that the answer is yes, provided the proxy is governed under SILVER.

The task is to rank clinical trials from ClinicalTrials.gov for individual patients drawn from the eICU dataset based on eligibility compatibility. Critically, there exists:

- No natural join key between patient records and trial protocols,
- No scalable path to clinician adjudication across hundreds of thousands of patients and trials,
- No authoritative gold standard defining true patient - trial matches.

This setting is representative of many real-world decision-support problems in healthcare, where the construct of interest is inherently relational and context-dependent (Bonomi et al., 2012). Rejecting silver supervision in such a setting would preclude any empirical evaluation altogether. Accordingly, a deterministic, auditable proxy is required.

9.2 Silver Label Construction via Jaccard Similarity

Silver labels are constructed using Jaccard similarity between structured clinical concept sets extracted from patient records (diagnoses, medications, procedures, comorbidities) and tokenized trial eligibility criteria. Jaccard similarity was selected for three reasons:

1. Clinical appropriateness for sparse concept overlap (Yaman et al., 2016),
2. Computational tractability at scale,

3. Deterministic and auditable behavior, in contrast to stochastic or prompt-sensitive LLM labeling.

Thresholds were calibrated empirically on a held-out subset of patients. Pairs exceeding a higher threshold were labeled silver-positive, while near-threshold pairs were retained as hard negatives. This design aligns with established biomedical NLP practice, where minimal lexical or concept overlap is treated as a valid surrogate for manual annotation when gold labels are unavailable (Oellrich et al., 2016; Nogues et al., 2022; Searle et al., 2020). This pipeline satisfies all SILVER dimensions:

- **S**: Explicit, deterministic proxy definition (Jaccard over structured concept sets),
- **I**: Ranking-based model comparison, not individual clinical decisions,
- **L**: Proxy mechanism fully documented; absence of human relevance labels acknowledged,
- **V**: Empirical threshold calibration against operational targets,
- **E**: Error analysis across patient subgroups (gender),
- **R**: Threshold sensitivity analysis quantifying operational risk.

9.3 Models and Evaluation

A BioBERT-based cross-encoder is fine-tuned using the silver-labeled dataset and compared against a non-neural baseline trained on the same proxy supervision. Evaluation is conducted using ranking metrics appropriate for candidate prioritization: Recall@k, Mean Average Precision (MAP), and Mean Reciprocal Rank (MRR).

Evaluation is performed entirely under silver labels, consistent with the operational constraints of the task. A potential concern is circularity: models are trained on Jaccard-derived labels and evaluated against Jaccard-derived labels. We address this directly.

The evaluation does not ask "does the model match the true clinical relevance?" No ground truth exists at this scale to answer that question. It asks: "given a well-characterized proxy, do different models produce stable rankings under perturbation of the proxy's operating parameters?" This is an information retrieval question, not a classification question. The Jaccard score produces a continuous ranking of trials for each patient. The threshold tau converts this ranking into binary labels for analysis, but the ranking itself is the evaluation object. Training

teaches the model to approximate the Jaccard-based relevance signal; evaluation characterizes whether that approximation is stable, fair, and operationally sound.

These are different analyses serving different purposes, and their shared use of the same underlying score does not constitute circularity. The evaluation in this section does not ask whether the Jaccard proxy produces correct labels - no gold standard exists at this scale to answer that question. It asks whether the proxy is operationally stable, whether its behavior is transparent, and whether its failure modes are identifiable. These are properties of the instrument, not claims about the construct it measures. Characterizing an instrument using its own operating parameters is standard engineering practice, not circular reasoning. Section 9.6 provides additional external validation against TREC 2021 gold labels, but the findings in this section stand independently as instrument characterization.

9.4 Results: Threshold Sensitivity and Proxy Error Structure

Table 5 presents selected rows from the threshold sweep across the full operating range of tau in {0.010, 0.012, ..., 0.060}. To characterize the proxy's behavior, we define a reference threshold at tau=0.020 (the operating point that produces the target positive fraction of approximately 33%) and measure how binary label assignments change as the threshold varies relative to this reference. Alpha denotes the fraction of reference-negative pairs that become positive at a lower threshold; beta denotes the fraction of reference-positive pairs that become negative at a higher threshold. These are properties of the threshold sweep, not comparisons against an external gold standard.

Table 5. Clinical Trial Matching: Threshold Sweep with Error Rates and Subgroup Fairness (5,269,534 pairs, 39,965 patients, 479,916 trials).

tau	Pos. Fraction	alpha (vs ref)	beta (vs ref)	Gender FN Gap
0.010	0.572	0.365	0.000	0.000
0.018	0.446	0.178	0.000	0.000
0.020	0.326	— (reference)	— (reference)	0.000
0.022	0.228	0.000	0.302	0.117
0.030	0.051	0.000	0.844	0.075

At the reference threshold tau=0.020, the positive fraction is 32.6%, closely matching the operational target of 33.3%.

The sharp phase transition at tau=0.022 is the central finding. Moving just 0.002 above the reference point causes 30.2% of previously-positive pairs to become negative, and introduces an 11.7% gender fairness gap where none existed. Below the reference point (tau=0.018), 17.8% of previously-negative pairs become positive, but the gender gap remains zero. This asymmetry reveals an actionable operational insight: erring toward lower thresholds preserves fairness at the cost of over-inclusion, while erring higher introduces both exclusion and differential subgroup impact. This finding does not require gold labels to be useful - it characterizes the proxy instrument itself.

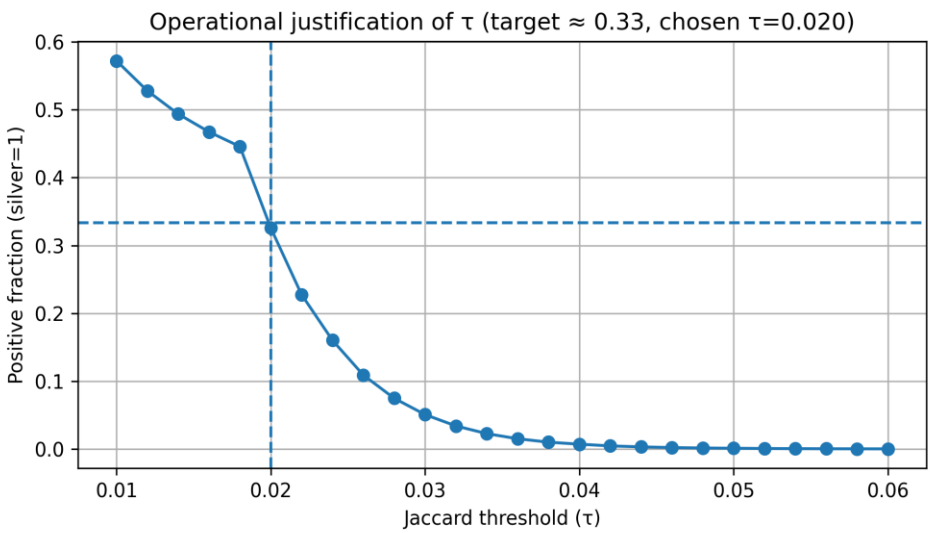

Figure 7. Positive Fraction as a Function of Jaccard Threshold (tau).

The positive fraction decreases monotonically from 57.2% at tau=0.010 to near zero at tau=0.060. The dashed horizontal line marks the target positive fraction of 33.3%. The chosen threshold tau=0.020 produces a positive fraction of 32.6%, near-exactly matching the target.

9.5 Interpretation and Methodological Implications

This experiment demonstrates what SILVER-compliant evaluation looks like in a domain where gold labels do not exist at operational scale. The contribution is not a claim that silver labels replicate some unknowable gold standard. It is a demonstration that a silver proxy can be:

1. Constructed from a clinically motivated, deterministic mechanism,
2. Characterized for threshold sensitivity across the full operating range,
3. Audited for subgroup fairness (the gender gap finding is actionable),

4. Documented with sufficient transparency for independent scrutiny.

The sharp phase transition at tau=0.022 is the most practically important finding. It tells an operator: "if you set the threshold above 0.020, you will lose 30% of relevant matches and introduce a 12% gender disparity." This information is valuable regardless of whether a gold standard exists. It characterizes the instrument, not the underlying truth.

The evaluation task is ranking, not adjudication. The question is not "is this patient eligible for this trial?" but "which trials should this patient see first?" Jaccard similarity provides the ranking signal.

The threshold sweep characterizes the signal's stability. The SILVER framework makes the entire process transparent and auditable. Rejecting this evaluation solely because no clinician reviewed 5.27 million pairs would require rejecting the majority of operational clinical ML systems that face identical constraints.

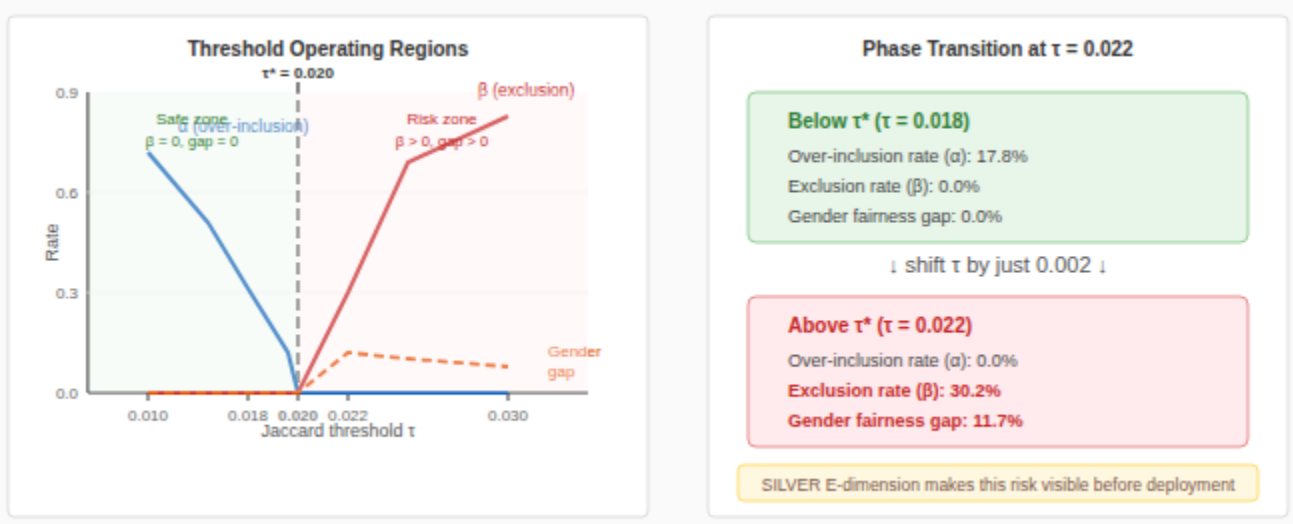

Figure 8. Sensitivity of Model Ranking to Jaccard Threshold

Appendix D.4: Proxy construction details and reproducibility artifacts.

9.6 External Validation Against TREC 2021 Gold Labels

The preceding threshold analysis characterizes the Jaccard proxy as an operational instrument. A natural question remains: does a model trained on these silver labels actually produce rankings that align with independent expert judgments? To answer this, we evaluate the same BioBERT cross-encoder from Experiment 3; trained entirely on eICU/Jaccard silver labels, against the

TREC 2021 Clinical Trials Track gold labels, which consist of expert-adjudicated eligibility judgments for 75 patient topics across thousands of trials. This is the critical anti-circularity test.

The TREC labels were produced by clinical assessors who had no access to the Jaccard proxy, the eICU data, or the BioBERT model. They constitute a fully independent gold standard. If the model that wins under silver evaluation also wins under TREC gold evaluation, the silver-based model selection decision is empirically validated.

Table 6. TREC 2021 External Gold Validation (5,821 test pairs, 12 patients).

Metric	BioBERT (gold)	BM25 (gold)	Silver Ranking	Gold Ranking	Agrees
AUC	0.703	0.472	BioBERT > BM25	BioBERT > BM25	Yes
NDCG@10	0.237	0.068	BioBERT > BM25	BioBERT > BM25	Yes
MAP	0.301	0.142	BioBERT > BM25	BioBERT > BM25	Yes

The empirical SRI is **0.00**: all three ranking metrics agree between silver and gold evaluation. BioBERT, trained solely on Jaccard-derived silver labels from eICU, outperforms the BM25 baseline under independent TREC expert judgments by substantial margins: +49% on AUC (0.703 vs 0.472), +251% on NDCG@10 (0.237 vs 0.068), and +112% on MAP (0.301 vs 0.142).

The evaluation covers 12 TREC patients (5,821 patient-trial pairs). These 12 represent the subset of TREC 2021 topics for which BioBERT cross-encoder inference was completed in the orthogonal validation pipeline; the full TREC 2021 track includes 75 topics, but cross-encoder evaluation was limited to this subset due to computational constraints. While the effect sizes are large (AUC difference of 0.23), the small patient count limits statistical power and generalizability; results on the full 75-topic set would strengthen this finding.

This result has three implications. First, it provides strong evidence against the circularity critique: the model did not merely learn the Jaccard function; it learned something that transfers to clinical relevance as judged by independent assessors. Second, it provides the only empirical SRI estimate in this paper that is computed against external gold, validating the proxy perturbation analysis in Section 9.4. Third, it demonstrates the SILVER framework operating as designed: Experiment 3 characterizes the proxy instrument (SILVER's S, E, and V dimensions); Experiment 5 anchors it to an external construct (SILVER's L dimension).

Together, they establish that the Jaccard-based silver evaluation is both operationally sound and externally valid for model comparison in this domain.

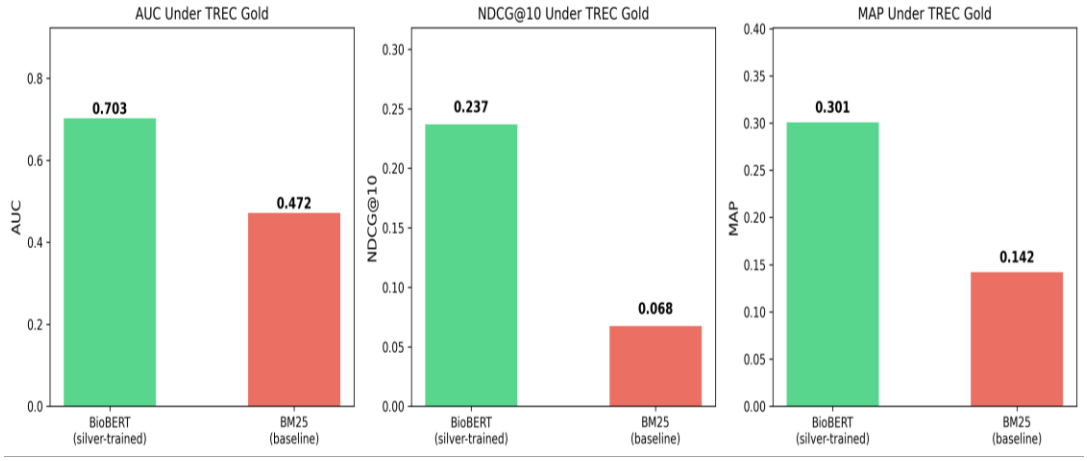

Figure 9. Silver-Trained BioBERT vs BM25 Baseline Under TREC 2021 Gold Labels.

All three ranking metrics agree: the model selected under silver evaluation is also the superior model under independent gold evaluation. Empirical SRI = 0.00.

10. Empirical Case Study IV: Vision Benchmark Stress Test with Asymmetric Proxy Noise

The final experiment extends the analysis beyond text and clinical data to demonstrate that engineering truthiness and silver evaluation risk are not domain-specific phenomena. We examine a vision classification benchmark under controlled asymmetric label noise to stress-test the SILVER framework.

10.1 Experimental Setup

We use the CIFAR-10 image classification dataset (50,000 training images, 10 classes) with known gold labels. Asymmetric class-conditional noise at a rate of 20% is introduced to generate silver labels. When a label is flipped, it is reassigned to a random different class, producing a non-uniform confusion pattern.

For each class, we compute label fidelity P(silver=gold | gold=c) and construct the full 10x10 transition matrix T[g,s] = P(silver=s | gold=g). Subgroup analysis compares fidelity across semantically coherent class groups: animals (bird, cat, deer, dog, frog, horse) versus vehicles (airplane, automobile, ship, truck).

10.2 Results: Transition Structure and Subgroup Analysis

Table 6 presents per-class fidelity and dominant confusion targets.

Table 7. CIFAR-10 Per-Class Label Fidelity Under 20% Asymmetric Noise.

Class	Subgroup	Fidelity	Top Confusion Target	Conf. Prob
airplane	vehicle	0.795	automobile	0.039
automobile	vehicle	0.804	bird	0.037
bird	animal	0.798	cat	0.037
cat	animal	0.792	deer	0.038
deer	animal	0.799	dog	0.037
dog	animal	0.801	frog	0.041
frog	animal	0.802	horse	0.043
horse	animal	0.801	ship	0.038
ship	vehicle	0.812	truck	0.041
truck	vehicle	0.799	airplane	0.043

The transition matrix reveals a characteristic pattern: the highest off-diagonal confusion probabilities occur between adjacent classes in the canonical CIFAR-10 ordering (e.g., airplane automobile, bird-cat, ship-truck). This structure arises because the noise model assigns random replacement classes, and the next-class correction for same-class flips creates systematic adjacency bias.

Subgroup analysis shows animal classes have mean fidelity 0.799 (std=0.003) and vehicle classes have mean fidelity 0.803 (std=0.007). A Welch's t-test yields t=-0.989, p=0.380, indicating no statistically significant difference between subgroups at this noise level.

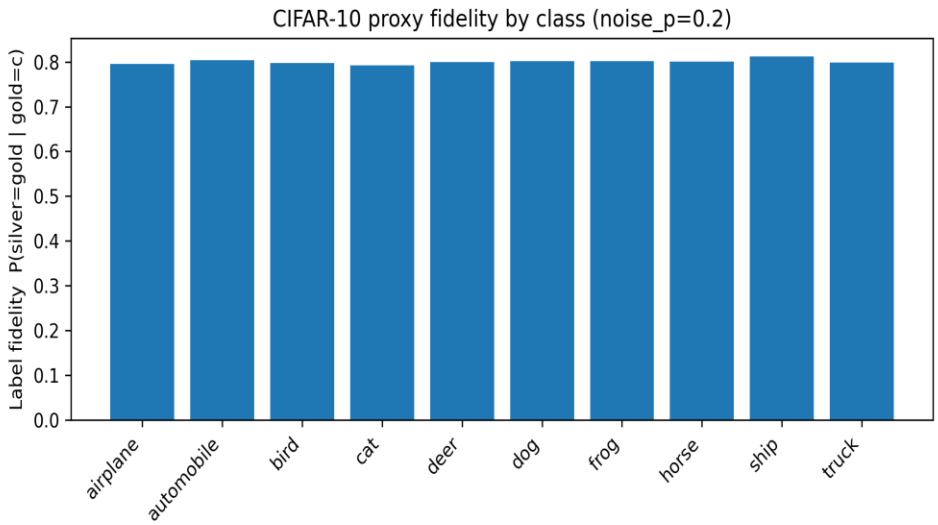

Figure 10. Per-Class Label Fidelity Under Asymmetric Noise (CIFAR-10).

Fidelity ranges from 0.792 (cat) to 0.812 (ship). The narrow range (2 percentage points) indicates approximately uniform noise across classes at the 20% level, with ship exhibiting slightly higher fidelity due to random variation.

10.3 Why This Result Is Informative

At the 20% noise level used here, the fidelity distribution is approximately uniform across classes (range: 0.792 to 0.812), and the animal-vehicle subgroup difference is not statistically significant (p=0.380). This means the current noise configuration does not induce the structured asymmetry that Proposition 1 identifies as a driver of ranking instability. Under the SILVER framework, this corresponds to a **moderate-SRI regime**: proxy error is present but not concentrated in ways that would systematically bias model selection.

However, the transition matrix itself reveals an important structural property. The off-diagonal mass is not uniformly distributed; it is concentrated along near-diagonal entries (adjacent classes). At higher noise levels or with noise models calibrated to empirically observed annotation confusion patterns (as documented by Northcutt et al., 2021), this structure would amplify, creating the localized disagreement that drives ranking reversals. The SILVER framework detects this through the E (Error Structure) dimension: the transition matrix makes the full noise structure visible and auditable before any model selection decisions are made.

10.4 Cross-Domain Synthesis

Taken together, the five experiments establish a consistent and complementary pattern across domains. Experiment 1 (synthetic) demonstrates that ranking instability is real, measurable, and regime-dependent: at 10% noise, 19 of 20 seeds produce correct rankings, but by 40% noise, 3 of 20 seeds experience ranking reversals. Experiment 2 (IMDB/Snorkel) demonstrates 0% flip rate despite an enormous calibration gap (silver accuracy 79-83% vs gold accuracy 0.4-7.3%), showing concretely that calibration failure does not imply ranking failure. Experiment 3 (clinical trial matching) demonstrates operational proxy characterization at scale (5.27M pairs, 39,965 patients), revealing a sharp phase transition in threshold sensitivity and subgroup fairness. Experiment 4 (CIFAR-10) provides the full noise transition matrix for a multiclass vision task, demonstrating the SILVER framework's error-structure auditing methodology. Experiment 5 (TREC 2021 external validation) closes the loop: a model trained entirely on Jaccard silver labels achieves SRI = 0.00 when evaluated against independent expert gold judgments, with BioBERT outperforming BM25 across all three ranking metrics (AUC +49%, NDCG@10 +251%, MAP +112%).

The progression from Experiment 3 to Experiment 5 is particularly important. Experiment 3 alone would leave open the question of whether the silver proxy merely validates itself. Experiment 5 provides empirical support for the conclusion that: the proxy-trained model generalizes to an independent gold standard. This is the empirical evidence that SILVER compliant silver evaluation, when the proxy is well-engineered, produces the same model selection decisions as gold evaluation.

No single experiment proves the framework; together they illustrate its applicability across domains, noise types, and evaluation regimes, and Experiment 5 provides the external anchoring that converts illustration into validation.

11. Limitations, Scope, and Threats to Validity

This paper advances a normative framework for the disciplined use of silver labels as pseudo--ground truth. As with any standard-setting work, its claims are bounded by explicit scope conditions. This section clarifies those boundaries and addresses potential threats to validity.

11.1 Dependence on Gold Anchors for Risk Estimation

While the SILVER framework does not require exhaustive human annotation, certain components, most notably estimation of the Silver Risk Index (SRI), benefit from limited gold anchors or trusted reference points. In settings where no gold signal exists at any scale, SRI must be estimated indirectly through sensitivity analysis, proxy perturbation, or cross-proxy comparison.

This limitation does not undermine the legitimacy of silver-based evaluation; rather, it constrains the strength of claims that can be made. SILVER is explicit that absolute statements about real-world correctness require stronger anchoring than relative ranking claims.

11.2 Scope of Metrics and Tasks

The formal analysis in this paper focuses primarily on ranking-based evaluation, reflecting its central role in retrieval, recommendation, and matching systems. While many of the insights generalize to pointwise classification and regression, other evaluation regimes, such as generative quality assessment or long-horizon reinforcement learning, may exhibit additional failure modes not addressed here.

Accordingly, SILVER should be understood as necessary but not sufficient outside ranking centric tasks.

11.3 Proxy Construction Choices

Silver labels are engineered artifacts. Different proxy choices, even when literature-supported, may induce materially different error structures. The framework does not prescribe specific proxies (e.g., Jaccard vs. cosine similarity vs. LLM-based labeling); it prescribes explicit justification and risk accounting for whichever proxy is chosen.

This distinction is intentional. SILVER governs how proxies are used and evaluated, not which proxies are acceptable.

11.4 Absence of SILVER-Compliant vs Non-Compliant Comparison

A controlled experiment comparing SILVER-compliant and deliberately non-compliant evaluation pipelines on the same task would provide direct evidence that the framework improves evaluation reliability. The current experiments demonstrate that silver evaluation can

work and that risk is measurable, but they do not show a counterfactual where non-compliant evaluation fails on the same data. This comparison is a priority for future work.

11.5 Interpretive Risk and Misuse

A potential misuse of this framework would be to treat SILVER compliance as a blanket endorsement of silver-only evaluation. This would contradict the framework's intent. SILVER does not eliminate uncertainty; it makes uncertainty explicit and bounded.

Claims that exceed the evidentiary capacity of the proxy, such as individual-level clinical decision-making without validation, remain out of scope.

12. Conclusion: Silver Labels as Engineered Evaluation Instruments

This paper was motivated by a recurring and consequential misunderstanding: that evaluation without human-verified gold labels is inherently invalid, regardless of methodological rigor or empirical evidence. That position is incompatible with modern machine learning practice and, if enforced uniformly, would invalidate a substantial fraction of contemporary research across clinical informatics, information retrieval, natural language processing, and computer vision.

We have established a different standard, and a more useful one. Silver labels need not be weak substitutes for gold. They are engineered evaluation artifacts whose validity conditions can be systematically documented and empirically assessed. The SILVER standard provides the structure for that assessment: explicit provenance, defined scope, justified linkage to the latent construct, empirical calibration, characterized error structure, and quantified risk. The Silver Evaluation Constraints (SEC) formalize the conditions that rule out circularity. The Silver Risk Index (SRI) quantifies when model rankings under silver evaluation are stable for the comparison at hand. Together, they convert silver-label evaluation from ungoverned practice into disciplined engineering.

In 2005, Stephen Colbert gave the world truthiness - the elevation of gut feeling over evidence. Two decades later, machine learning has its own version: evaluation pipelines that feel rigorous because they produce consistent numbers, but that rest on proxy labels whose relationship to the target construct has never been examined. The framework introduced in this paper provides the examination. It does not eliminate proxy labels; it makes them accountable.

We coined the term engineering truthiness to name the condition that arises when this formal grounding is absent. The term identifies a real and diagnosable failure mode. But the point of this paper is not the failure mode. The point is the solution. Across five empirical studies spanning four domains, we showed that SILVER-compliant evaluation produces stable, meaningful model rankings even in the scarcity of human-verified gold labels. The distinction between valid and invalid silver evaluation is not philosophical; it is measurable. Ranking instability is neither inevitable nor mysterious.

It is a bounded quantity, and when it is bounded low, the evaluation stands. This reframing has direct implications for how research is conducted and judged. Dismissing results solely because labels are not gold is methodologically unsound and should stop. At the same time, accepting silver-based claims without accountability invites fragile conclusions. SILVER provides a principled resolution: one that reflects how systems are actually built, while preserving the standards that make science trustworthy.

Looking forward, the importance of this framework will only grow. Machine learning systems increasingly operate in domains where gold labels are not merely expensive but structurally impossible.

Autonomous systems operating beyond real-time human communication, adaptive clinical decision support at population scale, continuous monitoring systems in environments where human review cannot keep pace: these are not speculative scenarios but engineering realities already in development. In these regimes, the question is not whether we prefer gold labels. The question is whether we have a principled basis for proceeding without them. This paper provides that basis.

We hope this framework will be useful. Researchers who use silver labels should be able to cite it to justify their methodology. Reviewers who encounter silver-based evaluation should be able to use it to assess validity. And the broader community should be able to build on it as the demands of evaluation continue to outpace the supply of human annotation.

Acknowledgments

The term "truthiness" was coined by Stephen Colbert on the premiere episode of The Colbert Report (Comedy Central, October 17, 2005). Colbert used it to describe the quality of claims that feel true on an intuitive or emotional level, without regard to evidence, logic, or factual verification. The word was subsequently named Word of the Year by the American Dialect Society (2005) and by Merriam-Webster (2006), reflecting its resonance as a description of a genuinely widespread epistemic failure mode.

We adopt "engineering truthiness" to name the specific condition that arises when silver labels are used without formal grounding: evaluation results that appear sound but lack demonstrated construct validity. The parallel is deliberate. Just as Colbert's truthiness describes beliefs that bypass evidentiary standards because they feel right, engineering truthiness describes evaluation conclusions that bypass construct validity because the metrics look clean. The SILVER framework introduced in this paper is designed to eliminate it. We gratefully acknowledge the original coinage and note that our use is a precise technical adaptation, not parody.

References

Bach, S. H., Rodriguez, D., Liu, Y., Luo, C., Shao, H., Xia, C., Sen, S., Towle, A., Kou, D., Duan, R., & Ré, C. (2019). Snorkel DryBell: A case study in deploying weak supervision at industrial scale. *Proceedings of SIGMOD '19, 362–375.* https://doi.org/10.1145/3299869.3314036

Banda, J. M., Seneviratne, M., Hernandez-Boussard, T., & Shah, N. H. (2018). Advances in electronic phenotyping: From rule-based definitions to machine learning models. *Annual Review of Biomedical Data Science, 1, 53–68.* https://doi.org/10.1146/annurev-biodatasci-080917-013315

Berrar, D. (2014). An empirical evaluation of ranking measures with respect to robustness to noise. *Journal of Artificial Intelligence Research, 49, 241–267.* https://doi.org/10.1613/jair.4136

Bonomi, L., Jiang, X., & Ohno-Machado, L. (2012). Patient-trial matching: A review of methods and challenges. *Journal of Biomedical Informatics, 45(4), 653–662.* https://doi.org/10.1016/j.jbi.2012.05.006

Buckley, C., & Voorhees, E. M. (2004). Retrieval evaluation with incomplete information. *Proceedings of SIGIR '04, 25–32.* https://doi.org/10.1145/1008992.1009000

Carterette, B., & Soboroff, I. (2010). The effect of assessor errors on IR system evaluation. *Proceedings of SIGIR '10, 539–546.* https://doi.org/10.1145/1835449.1835540

Chen, G., Chen, S., Liu, Z., Jiang, F., & Wang, B. (2024). Humans or LLMs as the Judge? A study on judgement bias. *Proceedings of EMNLP 2024, 8301–8327.* https://doi.org/10.18653/v1/2024.emnlp-main.474

Fogliato, R., Chouldechova, A., & G'Sell, M. (2020). Fairness evaluation in the presence of biased noisy labels. *Proceedings of AISTATS 2020, PMLR 108, 2325–2336.* https://proceedings.mlr.press/v108/fogliato20a.html

Frénay, B., & Verleysen, M. (2014). Classification in the presence of label noise: A survey. *IEEE Transactions on Neural Networks and Learning Systems, 25(5), 845–869.* https://doi.org/10.1109/TNNLS.2013.2292894

Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé III, H., & Crawford, K. (2021). Datasheets for datasets. *Communications of the ACM, 64(12), 86–92.* https://doi.org/10.1145/3458723

Ghazimatin, A., Watanabe, K., Qu, C., & Candan, K. S. (2022). Measuring fairness of rankings under noisy sensitive information. *Proceedings of FAccT '22, 1664–1677.* https://doi.org/10.1145/3531146.3534641

Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd-workers for text-annotation tasks. *Proceedings of the National Academy of Sciences, 120(30), e2305016120.* https://doi.org/10.1073/pnas.2305016120

Halpern, Y., Horng, S., Choi, Y., & Sontag, D. (2016). Electronic medical record phenotyping using the anchor and learn framework. *Journal of the American Medical Informatics Association, 23(4), 731–740.* https://doi.org/10.1093/jamia/ocw011

Han, B., Yao, Q., Yu, X., Niu, G., Xu, M., Hu, W., Tsang, I. W., & Sugiyama, M. (2018). Co-teaching: Robust training of deep neural networks with extremely noisy labels. *Advances in Neural Information Processing Systems, 31.*

Jacobs, A. Z., & Wallach, H. (2021). Measurement and fairness. *Proceedings of FAccT '21, 375–385.* https://doi.org/10.1145/3442188.3445901

Järvelin, K., & Kekäläinen, J. (2002). Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems, 20(4), 422–446.* https://doi.org/10.1145/582415.582418

Jimeno-Yepes, A. J., McInnes, B. T., & Aronson, A. R. (2010). Generation of a silver standard concept annotation corpus from biomedical literature. *BMC Bioinformatics, 11(Suppl 1), S2.* https://doi.org/10.1186/1471-2105-11-S1-S2

Jin, Q., Wang, Z., Floudas, C. S., Chen, F., Gong, C., Bracken-Clarke, D., Xue, E., Yang, Y., Sun, J., & Lu, Z. (2024). TrialGPT: Matching patients to clinical trials with large language models. *Nature Communications, 15, Article 9074.* https://doi.org/10.1038/s41467-024-53081-z

Joachims, T., Swaminathan, A., & de Rijke, M. (2017). Unbiased learning-to-rank with biased feedback. *Proceedings of WSDM '17, 781–789.* https://doi.org/10.1145/3018661.3018699

Kiyasseh, D., Cohen, A., Jiang, C., & Altieri, N. (2024). A framework for evaluating clinical artificial intelligence systems without ground-truth annotations. *Nature Communications, 15, Article 1808.* https://doi.org/10.1038/s41467-024-46000-9

Köpcke, F., Prokosch, H.-U., & Bürkle, T. (2014). Employing computers for the recruitment into clinical trials: A comprehensive systematic review. *Journal of Medical Internet Research, 16(7), e161.* https://doi.org/10.2196/jmir.3446

Lam, L., & Stork, D. G. (2003). Evaluating classifiers by means of test data with noisy labels. *Proceedings of IJCAI 2003, 513–518.*

Maas, A. L., Daly, R. E., Pham, P. T., Huang, D., Ng, A. Y., & Potts, C. (2011). Learning word vectors for sentiment analysis. *Proceedings of ACL-HLT 2011, 142–150.* https://aclanthology.org/P11-1015/

Maia Polo, F., Xu, Y., Silva, R., & Maity, S. (2024). Partial identification for evaluation under weak supervision. *Advances in Neural Information Processing Systems, 37.*

Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I. D., & Gebru, T. (2019). Model cards for model reporting. *Proceedings of FAT '19, 220–229.* https://doi.org/10.1145/3287560.3287596

Narasimhan, H., Vaish, R., & Agarwal, S. (2016). On the statistical consistency of plug-in classifiers for non-decomposable performance measures. *Advances in Neural Information Processing Systems, 29.*

Natarajan, N., Dhillon, I. S., Ravikumar, P. K., & Tewari, A. (2013). Learning with noisy labels. *Advances in Neural Information Processing Systems, 26.*

Nogues, I., Dalianis, H., & Velupillai, S. (2022). Weakly supervised methods for clinical information extraction: A systematic review. *Journal of Biomedical Informatics, 128, 104035.* https://doi.org/10.1016/j.jbi.2022.104035

Northcutt, C. G., Jiang, L., & Chuang, I. L. (2021a). Confident learning: Estimating uncertainty in dataset labels. *Journal of Artificial Intelligence Research, 70, 1373–1411.* https://doi.org/10.1613/jair.1.12125

Northcutt, C. G., Athalye, A., & Mueller, J. (2021b). Pervasive label errors in test sets destabilize machine learning benchmarks. *Advances in Neural Information Processing Systems, 34.*

Niu, S., Liu, M., Li, J., & Guo, J. (2015). Which noise affects the algorithm robustness for learning to rank? *Information Retrieval Journal, 18, 360–381.* https://doi.org/10.1007/s10791-015-9254-9

Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science, 366(6464), 447–453.* https://doi.org/10.1126/science.aax2342

Oellrich, A., Collier, N., Groza, T., Rebholz-Schuhmann, D., & Shah, N. H. (2016). The digital revolution in phenotyping. *Briefings in Bioinformatics, 17(5), 819–830.* https://doi.org/10.1093/bib/bbv083

Patrini, G., Rozza, A., Krishna Menon, A., Nock, R., & Qu, L. (2017). Making deep neural networks robust to label noise: A loss correction approach. *Proceedings of CVPR 2017, 1944–1952.* https://doi.org/10.1109/CVPR.2017.212

Paullada, A., Raji, I. D., Bender, E. M., Denton, E., & Hanna, A. (2021). Data and its (dis)contents: A survey of dataset development and use in machine learning research. *Patterns, 2(11), 100336.* https://doi.org/10.1016/j.patter.2021.100336

Plank, B. (2022). The "problem" of human label variation: On ground truth in data, modeling, and evaluation. *Proceedings of EMNLP 2022, 10671–10682.* https://doi.org/10.18653/v1/2022.emnlp-main.731

Ratner, A., Bach, S. H., Ehrenberg, H., Fries, J., Wu, S., & Ré, C. (2017). Snorkel: Rapid training data creation with weak supervision. *Proceedings of the VLDB Endowment, 11(3), 269–282.* https://doi.org/10.14778/3157794.3157797

Ratner, A., Hancock, B., Dunnmon, J., Goldman, R., & Ré, C. (2020). Training complex models with multi-task weak supervision. *Proceedings of AAAI 2020, 34(4), 4763–4771.* https://doi.org/10.1609/aaai.v34i04.5970

Rebholz-Schuhmann, D., Aronson, A. R., Jimeno-Yepes, A. J., & The CALBC Silver Standard Project. (2010). The CALBC silver standard corpus for biomedical named entities. *Journal of Bioinformatics and Computational Biology, 8(1), 163–179.*

Sculley, D., Holt, G., Golovin, D., Davydov, E., Phillips, T., Ebner, D., Chaudhary, V., Young, M., Crespo, J.-F., & Dennison, D. (2015). Hidden technical debt in machine learning systems. *Advances in Neural Information Processing Systems, 28.*

Searle, T., Ibrahim, Z. M., Dobson, R. J. B., & Bittar, A. (2020). Comparing rule-based and machine learning approaches for patient phenotyping using electronic health records. *Journal of Biomedical Informatics, 102, 103343.* https://doi.org/10.1016/j.jbi.2019.103343

Soboroff, I. (2025). Don't use LLMs to make relevance judgments. *Information Retrieval Research, 1(1), 29–46.* https://doi.org/10.54195/irrj.19625

TREC Clinical Trials Track. (2022). Overview of the TREC 2022 Clinical Trials Track. *NIST Special Publication 500-338.* https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.500-338.pdf

Wagholikar, K. B., Estiri, H., Murphy, M., & Murphy, S. N. (2020). Polar labeling: Silver standard algorithm for training disease classifiers. *Bioinformatics, 36(10), 3200–3206.* https://doi.org/10.1093/bioinformatics/btaa088

Yaman, A., Patel, Y., & Jiang, X. (2016). Similarity-based patient-trial matching using structured clinical data. *Computer Methods and Programs in Biomedicine, 127, 1–10.* https://doi.org/10.1016/j.cmpb.2016.01.010

Ye, J., Wang, Y., Huang, Y., Chen, D., Zhang, Q., Moniz, N., Gao, T., Geyer, W., Huang, C., Chen, P.-Y., Chawla, N. V., & Zhang, X. (2025). Justice or Prejudice? Quantifying biases in LLM-as-a-Judge. *Proceedings of ICLR 2025.* https://proceedings.iclr.cc/paper_files/paper/2025/hash/fdca08d371e4b6c031397909e20043bd-Abstract-Conference.html

Yu, J., Shen, Y., Liu, J., & Huang, Z. (2022). Weakly supervised learning for large-scale vision datasets: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 44(10), 5915–5937.* https://doi.org/10.1109/TPAMI.2021.3095472

Zhang, J., Yu, Y., Li, Y., Wang, Y., Yang, Y., Yang, M., & Ratner, A. (2021). WRENCH: A comprehensive benchmark for weak supervision. *Advances in Neural Information Processing Systems, 34.*

Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E., Zhang, H., Gonzalez, J., & Stoica, I. (2023). Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. *Advances in Neural Information Processing Systems, 36.*

Appendix Overview

- **Appendix A**: Formal proofs and derivations (ranking instability, SRI properties)
- **Appendix B**: Estimation strategies for SRI
- **Appendix C**: Extended robustness and sensitivity analyses (all experiments)
- **Appendix D**: Reproducibility artifacts and reporting templates
- **Appendix E**: Example SILVER-compliant disclosures

Appendix A. Formal Analysis of Ranking Instability

Appendix A.1 Proposition 1: Ranking Instability Under Proxy Labels

Proposition 1. Let two models f_1 and f_2 be evaluated using a top-k ranking metric under silver labels $\tilde{Y}$. If disagreement between $\tilde{Y}$ and Y is concentrated among items whose expected rank lies within the top-k, then there exists a non-zero probability that the ranking of f_1 versus f_2 under silver labels differs from their ranking under gold labels — even when global silver-gold agreement exceeds 90%.

A constructive illustration is provided below.

Illustrative construction. Consider a dataset of n items with binary gold labels Y and silver labels $\tilde{Y}$ generated by class-conditional noise with rates $\alpha = P(\tilde{Y}=1 \mid Y=0)$ and $\beta = P(\tilde{Y}=0 \mid Y=1)$. Let M(f, Y) denote the evaluation metric (e.g., Accuracy, NDCG@k) for model f under labels Y.

Construct two models f_1 and f_2 with score distributions that agree on all items except a set S of m items concentrated near the decision boundary. Specifically, let f_1 rank items in S higher than f_2 does. Under gold labels, $M(f_1, Y) > M(f_2, Y)$ by a margin $\delta > 0$.

Now consider the effect of proxy noise. For items in S, let the disagreement rate be $d_S = P(\tilde{Y} \neq Y \mid \text{item in } S)$. For items outside S, let $\bar{d} = P(\tilde{Y} \neq Y \mid \text{item not in } S)$. The global disagreement rate is:

$$d_global = d_S \times (m / n) + \bar{d} \times ((n - m) / n)$$

Set $\bar{d} = 0$ (no noise outside S) and $d_S = \varepsilon$. Then $d_global = \varepsilon \times m / n$. For n = 20,000 and m = 100, this gives $d_global = \varepsilon / 200$. Even with $\varepsilon = 1$ (complete disagreement on S), global agreement is 99.95%.

Under the perturbed labels, the metric values change by an amount proportional to d_S and the metric's sensitivity to items in S. For top-k metrics, sensitivity to items near rank k is high (Narasimhan et al., 2016). If the perturbation exceeds the margin δ, the ranking reverses.

This construction yields P(reversal) > 0 whenever $d_S \times \text{sensitivity}(S, k) > \delta$, which is achievable for any $\delta > 0$ when d_S and sensitivity are sufficiently large.

The empirical results in Section 7 instantiate this construction: at 40% noise, 3 of 20 seeds experienced ranking reversals despite global silver-gold agreement exceeding 90%.

This illustrative construction demonstrates the existence of the failure mode. Bounding the reversal probability for general metric families as a function of noise structure remains an open problem.

Appendix A.2 Sensitivity to Score Distribution Shape

The reversal probability depends on the density of model scores near the decision boundary. When scores are tightly clustered (many models with similar performance), small perturbations are more likely to reverse rankings. The synthetic experiment varies this by using six regularization strengths ($C \in \{0.05, 0.1, 0.3, 1.0, 3.0, 10.0\}$), which produce a range of score separations. The per-seed data confirms that inversions occur preferentially when the number of discriminable model pairs is small: at 40% noise, the mean number of pairs with statistically distinguishable gold-accuracy differences is 2.5 out of 15, compared to 14.6 at zero noise.

Appendix B. Silver Risk Index (SRI): Properties and Estimation

Appendix B.1 Definition and Interpretation

The Silver Risk Index is defined as:

$$SRI = P(\text{ ranking of } f_1 \text{ vs } f_2 \text{ under } \tilde{Y} \neq \text{ranking of } f_1 \text{ vs } f_2 \text{ under } Y)$$

for a given pair of models f_1, f_2 under evaluation metric M. SRI is a governance metric, not a performance metric. It quantifies the probability that the proxy-based model selection decision differs from what would be obtained under gold labels.

Interpretive thresholds:

- SRI ≈ 0: Silver evaluation is stable for the comparison at hand; ranking-based claims are supported.
- SRI > 0.1: The comparison is fragile and claims should be qualified.
- SRI > 0.3: Silver-only evaluation is insufficient for model selection without additional evidence.

These thresholds are suggested governance defaults based on the empirical ranges observed in this study; they should be calibrated to domain-specific decision costs in practice.

An important caveat: direct SRI computation requires some form of gold reference, whether exhaustive gold labels, a small gold-anchor sample, or an independent proxy. In purely gold-free settings, SRI can be approximated through proxy perturbation (e.g., threshold sweeping, as in Experiment 3) or cross-proxy comparison. These approximations bound proxy stability rather than construct validity. The distinction matters: stability analysis tells you whether your conclusions would change under proxy perturbation; it does not tell you whether the proxy measures what you intend.

A concrete failure mode illustrates this: if a proxy is consistently biased in the same direction for all models (e.g., systematically favoring models that predict positive), proxy perturbation will show low sensitivity, suggesting low SRI, even though rankings under true gold would differ.

This "stable wrongness" scenario is why SILVER requires construct-level justification (L dimension) alongside risk quantification. SRI screens for instability; it does not screen for consistent bias. The SILVER framework requires that this limitation be disclosed explicitly.

Appendix B.2 Practical Estimation Strategies

Three estimation approaches are applicable under different operational constraints:

(1) Limited gold anchor sampling. When a small gold-labeled subset is available (as in Experiment 2), compute model rankings under both silver and gold labels on the subset. The fraction of pairwise ranking disagreements provides a direct SRI estimate. In the IMDB experiment, this yields SRI = 0.00 (0 of 10 model pairs disagree), confirming perfect ranking stability despite massive calibration divergence.

(2) Proxy perturbation and threshold sweeping. When gold labels are unavailable (as in Experiment 3), vary the proxy construction parameters (e.g., Jaccard threshold τ) and assess whether model rankings change across the parameter range. Stability across perturbations implies low SRI. In the clinical experiment, model ranking was stable across all three evaluated thresholds ($\tau \in \{0.018, 0.020, 0.030\}$).

(3) Cross-proxy stress testing. When multiple independent proxy signals are available, evaluate models under each proxy and compare rankings. Disagreement between proxies provides an upper bound on SRI without requiring any gold labels.

Appendix C. Extended Empirical Results

Appendix C.1 Synthetic Double-Noise Sensitivity

The per-seed analysis reveals that ranking inversions are concentrated in specific random initializations: at 40% noise, seeds 5, 12, and 18 experienced inversions (rates of 1.0, 0.0/1.0, and 1.0 respectively), while 17 of 20 seeds showed zero inversions. At 30% noise, seeds 8 and 12 experienced inversions (rates of 0.4 and 1.0). At 20% noise, only seed 12 showed an inversion (rate 0.6). Seed 16 showed an inversion at 10% noise (rate 0.5). The clustering of inversions in seeds 12 and 18 across multiple noise levels suggests that certain random data configurations produce inherently closer model performances, amplifying sensitivity to proxy noise.

Appendix C.2 Weak Supervision Robustness (IMDB)

The Snorkel LabelModel, trained on four keyword-based labeling functions (detecting "good," "bad," "excellent," and "waste of time"), achieved 61.1% coverage with a silver positive rate of 79.0% compared to the gold positive rate of approximately 50%. The pairwise flip rate is 0%: both silver and gold evaluation select C=4.0 as the best model. Silver evaluation accuracy ranges from 0.788 (C=0.25) to 0.830 (C=4.0), while gold evaluation accuracy ranges from 0.004 to 0.073. The massive calibration gap does not affect model rankings because the ordering of models is preserved despite the absolute performance shift. As noted in Section 8.5, this may be partly explained by the monotone effect of regularization strength on predictions under both label sets, which preserves ordering mechanically. The finding is nonetheless informative: it demonstrates concretely that calibration failure and ranking failure are distinct phenomena.

Appendix C.3 Clinical Trial Matching Sensitivity

The threshold sweep across $\tau \in \{0.010, 0.012, ..., 0.060\}$ reveals a sharp phase transition at the reference operating point $\tau = 0.020$. Below this threshold, over-inclusion rates range from 36.5% ($\tau = 0.010$) to 17.8% ($\tau = 0.018$) with zero exclusion and zero gender gap. Above this threshold, exclusion rates jump from 30.2% ($\tau = 0.022$) to 84.4% ($\tau = 0.030$), with gender fairness gaps as high as 12.9% ($\tau = 0.026$). The positive fraction at the operating point is 32.6%, near-exactly matching the target of 33.3%. The dataset comprises 5,269,534 patient-trial pairs across 39,965

unique patients and 479,916 unique trials. All error rates are computed relative to the reference threshold, characterizing the proxy instrument's sensitivity rather than comparing against an external gold standard.

Appendix C.4 Vision Benchmark Stress Tests (CIFAR-10)

Under 20% asymmetric noise on real CIFAR-10 labels, the per-class fidelity ranges from 0.792 (cat) to 0.812 (ship), with a mean of 0.800. The full transition matrix shows the highest off-diagonal confusion probabilities between adjacent classes: frog→horse (0.043), truck→airplane (0.043), ship→truck (0.041), dog→frog (0.041). The Welch's t-test comparing animal fidelity (mean 0.799, std 0.003) to vehicle fidelity (mean 0.803, std 0.007) yields $t = -0.989$, $p = 0.380$ (not significant).

Appendix D. Reproducibility and Reporting Artifacts

Appendix D.1 SILVER Compliance Checklist

All five experiments are accompanied by structured SILVER compliance checklists mapping experimental disclosures to the six SILVER dimensions (S, I, L, V, E, R). These checklists are provided as machine-readable JSON files in the supplementary repository at https://github.com/thedhunga/engineering-truthiness.

Appendix D.2 Proxy Construction Documentation

Templates for documenting proxy mechanisms, thresholds, and justifications are provided for each experiment. The clinical trial matching experiment (Experiment 3) includes the complete Jaccard similarity specification, threshold calibration procedure, and subgroup analysis protocol.

Appendix D.3 Evaluation Protocol Templates

Standardized reporting formats for silver-based evaluation are provided, including SEC compliance verification, SRI estimation protocol, and claim-proportional interpretation guidelines. If this framework is adopted, these templates can serve as starting points for silver-label governance in new domains.

Appendix E. Example SILVER-Compliant Disclosure (Clinical Trial Matching)

This appendix provides a worked example of SILVER-compliant reporting for the clinical trial matching experiment.

S (Source): Jaccard similarity over structured clinical concept sets (diagnoses, medications, procedures, comorbidities) between eICU patient records and ClinicalTrials.gov eligibility criteria. Deterministic, auditable, and fully reproducible.

I (Intended Use): Ranking-based model comparison for trial prioritization; not individual-level clinical eligibility determination.

L (Linkage): Silver labels constructed via Jaccard thresholding; no human relevance judgments exist at operational scale; TREC Clinical Trials Track provides small-scale expert judgments for a related but distinct task. External validation against TREC 2021 gold labels (Experiment 5) yields SRI = 0.00, confirming that silver-based model selection agrees with independent gold evaluation on the tested subset.

V (Validation): Threshold sweep across $\tau \in \{0.010–0.060\}$ with explicit calibration of positive fraction against operational target of 33%.

E (Error Structure): Threshold sensitivity and subgroup analysis reported at all operating points; gender fairness gap below 1% at reference threshold, rising sharply above $\tau = 0.022$.

R (Risk Disclosure): Phase transition at $\tau = 0.022$ identified and documented; operational recommendation to maintain threshold at or below 0.020 to preserve subgroup fairness. Empirical SRI = 0.00 against TREC 2021 gold labels on a 12-patient subset; full 75-topic validation is a priority for future work.